

ERROR MEMORY: RETRIEVAL-AUGMENTED CORRECTION OF TIME-SERIES FORECASTS

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ABSTRACT

Retrieval augmentation is an attractive way to expose a trained time-series forecaster to rare historical conditions without touching its weights, and existing methods retrieve on the *input*: they find past windows whose context resembles the query and reuse the futures that followed. We argue that this is the wrong retrieval object. A modern forecaster already predicts the routine level, phase, and rhythm of a series; what it repeatedly misses is a small, recurring, and *model-specific* residual, which whole-window similarity does not isolate and a borrowed historical future does not preserve. We introduce TIMERAf, which retrieves a trained forecaster’s own past errors. It stores chronologically observed validation residuals, queries them with a descriptor built from both the observed context and the forecaster’s intended future, shrinks the retrieved correction where neighbouring errors disagree, and applies it only when validation prefers it to a simpler correction or to leaving the forecast untouched. Retrieving a future is not a separate mechanism: anchored to the query, it decomposes exactly into the retrieved error plus the drift between the two forecast trajectories, so TIMERAf exposes that drift as one weight and lets validation choose the retrieval object per deployment. In a 585-setting comparison where every system corrects identical base forecasts, TIMERAf improves all reported metrics in 541 settings, against 490 for pure error retrieval, 478 for a strong endpoint-aligned analog-future control, and 384 and 375 for the published SARAF and RAFT retrieval operators, and takes the best average error in every task family; the selected weight is non-zero in 435 settings and partial rather than a full overwrite in 299 of them. TIMERAf improves every reported metric in 562/585 settings spanning 13 architectures, 16 datasets, and horizons from 6 to 720 steps, and in 81/94 settings spanning static ensembles, AutoML, and a forecasting foundation model, where error retrieval is the selected correction in 70 settings.

1 INTRODUCTION

Time-series forecasting supports decisions in electricity markets, traffic management, and industrial operations, where the series mixes seasonality, trend, abrupt change, and multiscale dependence (Chen et al., 2001; Lago et al., 2021). A decade of architectural work has attacked this mixture through frequency decomposition, patching, cross-variable attention, multiscale mixing, exogenous conditioning, and large-corpus pretraining (Zhou et al., 2022; Nie et al., 2023; Wu et al., 2023; Liu et al., 2024b; Wang et al., 2024a;c; Ansari et al., 2024). The resulting models are strong at what they were designed for: they reproduce the routine level, phase, and rhythm of a series with small average error.

What such a model gets wrong is therefore not the routine part of the signal. The residual left after a trained forecaster has done its work is a small but structured signal that concentrates on infrequent events, turning points, and context-dependent biases, and that recurs whenever the generating condition recurs. Retrieval exploits recurrence without touching a deployed model’s weights, and a growing family of retrieval-augmented forecasters does exactly this (Tire et al., 2024; Zhang et al., 2024; Han et al., 2025; Ning et al., 2025). Almost all of them retrieve on the *input*: they score historical windows by similarity to the query

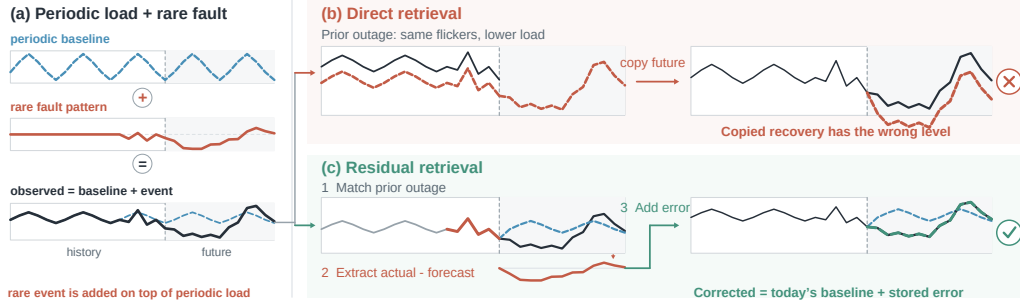


Figure 1: **Retrieve the error, not the input.** (a) An observed series is the sum of a dominant periodic component and an infrequent event. (b) Retrieving on the input matches a past window’s *routine* structure, which carries almost all of the window’s energy; copying the future that followed then injects the old level and phase on top of a forecast that already had them right. (c) Subtracting the forecaster’s own prediction from that past outcome leaves only what it missed, turning the retrieved quantity into a correction that can be added to today’s forecast.

context and reuse the futures that followed. Figure 1 shows why that object fits the error poorly. Whole-window similarity is dominated by the periodic and trend structure the base forecast already predicts, so the short prefix announcing a rare event rarely decides which neighbours are returned; a well-matched neighbour’s future still carries its own level, phase, and covariate response, so transferring it overwrites parts of the forecast that were already correct; and, beyond what the figure shows, the retrieved neighbourhood is identical no matter which forecaster is being augmented, so it cannot adapt to that forecaster’s blind spot. Hence our central question:

When we augment a time-series forecaster with retrieval, what should we retrieve?

Our answer is **TIMERAF**, a post-hoc correction layer that retrieves a trained forecaster’s **own past errors**. It builds on the model-relative decomposition $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{r}$, which splits a realized future into what a specific forecaster predicted and what it missed: rather than replacing $\hat{\mathbf{y}}$, **TIMERAF** keeps it as the query-specific prior and retrieves $\mathbf{r}$ from a chronological memory of validation residuals whose targets were already observable. Two choices make the retrieval model-adaptive. The query descriptor is built from the observed context *and* the forecaster’s intended future, so one window is described differently depending on what the model predicts for it, and the memory stores that model’s errors, so a different backbone retrieves a different correction from the same data. Since retrieved errors help only when errors recur, **TIMERAF** shrinks a correction whose neighbours disagree and applies it only when validation prefers it to simpler corrections or to leaving the forecast untouched. The two objects then turn out not to be rivals: anchoring a retrieved future to the query and subtracting the base forecast splits it exactly into the retrieved error plus the drift between the two forecast trajectories, so input retrieval is error retrieval with that drift forced on. **TIMERAF** exposes the drift weight instead, so the residual stays in every correction while validation decides how much of a neighbour’s trajectory to borrow on top of it.

Our contributions are threefold. (i) **A shift of retrieval object:** we identify the retrieval object, rather than the retrieval architecture, as what decides whether retrieval augmentation helps a trained forecaster, and show that retrieving a future and retrieving an error are two ends of one additive correction whose error end is the right default. (ii) **A practical algorithm:** **TIMERAF** turns that argument into a chronologically causal, validation-gated correction layer that selects its own retrieval object and needs neither retraining nor a learned retriever. Because it consumes only predictions and observed targets, never gradients or weights, one interface covers neural architectures, static ensembles, AutoML systems, and foundation models. (iii) **A systematic study:** we compare retrieval objects under a pre-registered protocol that gives every system identical base forecasts, including the operators of two published retrieval-augmented forecasters, and measure the layer across 13

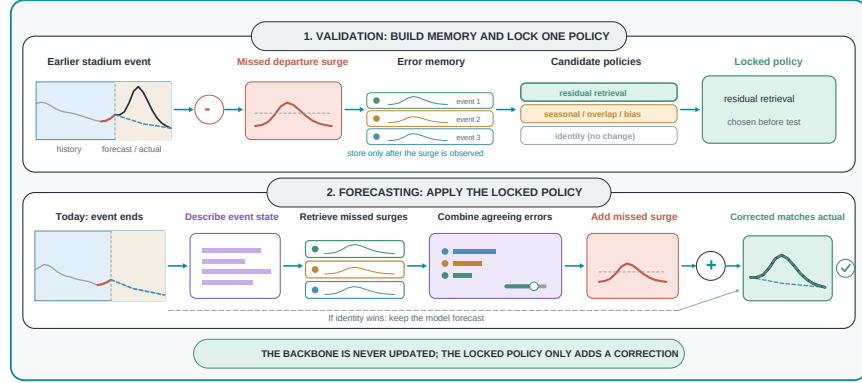


Figure 2: **TimeRaf separates validation-time calibration from forecast-time correction.** On a held-out validation split, the forecaster’s own predictions are differenced against realized targets to build an error memory, and one correction policy is locked. At forecast time the backbone is never updated: the locked policy either retrieves agreeing past errors and adds them to the current forecast, or falls back to a simpler correction, or leaves the forecast unchanged. The actual continuation is never available at forecast time.

architectures, six further systems, and a prequential study. It improves every reported metric in 541 of 585 settings with the base forecast held fixed, and in 96.1% of them when each backbone is corrected end to end.

2 PRELIMINARIES

Notation. For multivariate forecasting with d variables, the task at origin t is to predict the next H steps from the most recent L observations. Let $\mathbf{x}_t \in \mathbb{R}^{L \times d}$ be the observed context, $\mathbf{y}_t \in \mathbb{R}^{H \times d}$ its realized future, and $f: \mathbb{R}^{L \times d} \rightarrow \mathbb{R}^{H \times d}$ a forecaster with prediction $\hat{\mathbf{y}}_t = f(\mathbf{x}_t)$. We assume f has already been trained and deployed: we may query it, but its parameters are never updated. The quantity this work centres on is the *model-relative residual* $\mathbf{r}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$, defined jointly by the data and the particular forecaster and observable only after the whole target window has elapsed. A retrieval-augmented forecaster additionally has a memory $\mathcal{M}_t$ of past items and emits a corrected forecast $\tilde{\mathbf{y}}_t$; methods differ in whether those items hold prior contexts with the futures that followed or, as we propose, the same forecaster’s prior residuals. Throughout, *causal* means free of lookahead: since $\mathbf{r}_i$ needs the complete future at origin i , an item may enter $\mathcal{M}_t$ only if $i + H < t$.

Problem 1 (Retrieval-augmented correction of a trained forecast) *Given a trained forecaster f , a query context $\mathbf{x}_t$ with base forecast $\hat{\mathbf{y}}_t = f(\mathbf{x}_t)$, and a memory $\mathcal{M}_t$ restricted to items whose targets were fully observed before t , construct an additive correction c_t so that $\tilde{\mathbf{y}}_t = \hat{\mathbf{y}}_t + c_t$ is more accurate than $\hat{\mathbf{y}}_t$ with respect to $\mathbf{y}_t$, without updating f and without observing any part of $\mathbf{y}_t$.*

Problem 1 isolates the question of Section 1: the base forecast, the data, and the information boundary are fixed, so any accuracy difference is attributable to what $\mathcal{M}_t$ stores and how it is used.

3 METHOD

TIMERAF has three components, sketched in Figure 2: a chronological memory of a trained forecaster’s own errors (Section 3.1), a model-conditioned retriever that turns them into a reliability-shrunk correction whose *object* is itself tunable (Section 3.2), and a validation gate that decides whether to apply it (Section 3.3).

3.1 RETRIEVING ERRORS INSTEAD OF INPUTS

Why the error is the right default object. The dominant design scores past windows by context similarity and reuses the futures that followed (Tire et al., 2024; Han et al., 2025; Lee et al., 2026; Kang et al., 2026; Zhou et al., 2026). Storing $\mathbf{r}_i$ instead of $\mathbf{y}_i$ fixes three properties of that object. Raw-window distance is dominated by the periodic and trend energy the base forecast already predicts, so the short prefix announcing an unusual continuation rarely decides the neighbourhood, whereas a residual has that explained component subtracted out. A retrieved future is an absolute trajectory carrying its own level, phase, and covariate response, so injecting it overwrites parts of $\hat{\mathbf{y}}_t$ that were already correct, while a residual can only move the forecast. And an input neighbourhood is identical for every forecaster, whereas memory indexed to one forecaster suppresses the correction when that backbone has no recurring error. Appendix Figure 4 shows all three effects on measured ETTh1 forecasts: neighbours with matching contexts have futures at visibly different levels, but their residuals share one error shape.

The two objects are one family. Retrieving a future and retrieving an error are not alternatives; they are the two ends of a single additive correction. Anchoring neighbour i ’s future to the query endpoint, as input retrieval does, and subtracting the base forecast gives an exact decomposition,

$$(\mathbf{y}_i + \delta_i) - \hat{\mathbf{y}}_t = \mathbf{r}_i + \mathbf{g}_i, \quad \mathbf{g}_i = \hat{\mathbf{y}}_i + \delta_i - \hat{\mathbf{y}}_t, \quad \delta_i = \mathbf{x}_{t,L} - \mathbf{x}_{i,L}, \quad (1)$$

where $\mathbf{x}_{t,L}$ is the final context step. The left side is what input retrieval adds to the forecast and $\mathbf{r}_i$ is what error retrieval adds; they differ by the drift $\mathbf{g}_i$ between the forecaster’s trajectory in the neighbour’s state and in the query’s. Input retrieval hard-codes a full drift term and prior error retrieval hard-codes none, and neither can be right everywhere, because $\mathbf{g}_i$ is exactly the part that overwrites level, phase, and rhythm. Section 3.2 therefore exposes its weight instead of fixing it.

Chronological error memory. Memory is built from predictions on a held-out validation split rather than from training data, whose residuals are optimistically biased by the fit that produced them. Each item stores the context, base forecast, calendar marks, and residual, and is admitted only after its complete target has elapsed; during calibration an earlier validation prefix supplies memory and a later suffix supplies queries, so parameters are never chosen on the origins that memory explains.

3.2 MODEL-CONDITIONED RETRIEVAL WITH RELIABILITY SHRINKAGE

Context–forecast descriptor. Retrieval needs a representation of the *state the forecaster is in*, not only the state the series is in. TIMERAf therefore concatenates normalized shapes, first differences, summary statistics, slopes, and calendar marks over the last $\ell \in \{48, L\}$ context steps *and* over the base forecast $\hat{\mathbf{y}}$ itself. Including $\hat{\mathbf{y}}$ is what makes the index model-conditioned: two backbones seeing one context but intending different futures land in different neighbourhoods. High-dimensional series are pooled into at most 16 contiguous channel groups, and features are standardized with memory statistics and ℓ_2 -normalized. The construction needs no learned encoder, so one implementation applies unchanged to every forecaster we study.

Retrieval, aggregation, and shrinkage. Cosine similarity selects the top- k eligible items and a temperature-scaled softmax turns similarities into weights:

$$s_{ti} = \phi_t^\top \phi_i, \quad w_{ti} = \frac{\exp((s_{ti} - s_{t,\max})/\tau)}{\sum_{j \in \mathcal{N}_k(t)} \exp((s_{tj} - s_{t,\max})/\tau)}, \quad \boldsymbol{\mu}_t = \sum_{i \in \mathcal{N}_k(t)} w_{ti} \mathbf{r}_i, \quad (2)$$

so the weighted mean $\boldsymbol{\mu}_t$ estimates the recurring error at origin t . That mean is trustworthy only where the retrieved errors agree, and agreement varies across the horizon and across channels: a backbone may carry a stable bias at short lead times and none at long ones. TIMERAf therefore computes a weighted variance σ_t^2 and effective neighbour count $k_{\text{eff}} =$

$(\sum_i w_{ti}^2)^{-1}$ per horizon step and channel group, and shrinks each component by its own signal-to-noise ratio:

$$\boldsymbol{\rho}_t = \frac{\boldsymbol{\mu}_t^2}{\boldsymbol{\mu}_t^2 + \lambda \sigma_t^2 / k_{\text{eff}} + \epsilon}, \quad c_t^{\text{ret}} = \alpha \boldsymbol{\rho}_t \odot \boldsymbol{\mu}_t, \quad \epsilon = 10^{-8}. \quad (3)$$

Components on which neighbours agree keep $\boldsymbol{\rho}_t$ near one; a small mean with large disagreement is attenuated toward zero, returning that component to the original prediction. Scale α and shrinkage strength λ come from validation (Appendix C).

One dial over the retrieval object. The same neighbourhood and the same weights also aggregate the drift term of Equation 1, which lets a single expression interpolate between the two objects:

$$c_t = \alpha (\boldsymbol{\rho}_t \odot \boldsymbol{\mu}_t + \beta \boldsymbol{\gamma} \odot \sum_{i \in \mathcal{N}_k(t)} w_{ti} \mathbf{g}_i), \quad \beta \in [0, 1], \quad (4)$$

where $\boldsymbol{\gamma} \in \mathbb{R}^H$ is a horizon profile, flat or rising linearly with lead time. Setting $\beta = 0$ recovers pure error retrieval, and $\beta = 1$ with $\alpha = 1$, a flat profile, and no shrinkage reproduces the endpoint-aligned analogue mean that input-retrieval systems emit, so both published objects are literal special cases. Intermediate β is the case neither can express: keep the base trajectory but pull it partway towards the neighbourhood’s. It costs one extra weighted mean over the same index, and $\mathbf{g}_i$ vanishes on its own when the forecaster is accurate and its neighbours match, which is exactly the regime where overwriting would hurt most.

3.3 VALIDATION-GATED CORRECTION

Why a gate is necessary. Error retrieval rests on a testable assumption: that this forecaster’s errors recur in retrievably similar states. Where they do not, the right correction is a simpler one or none at all, and which case holds varies sharply across backbones and datasets. Besides the retrieved correction, the candidate pool therefore holds **identity** ($c_t = 0$), a **seasonal blend** repeating a recent period, an **overlap blend** of earlier forecasts of the same target with age decay, a **static bias** from the mean validation residual, and a **causal rolling bias** over recent elapsed residuals (Appendix C). The last two share error memory with retrieval but discard its state-conditioning, which is what makes the comparison informative.

Selection rule. For the metric set $\mathcal{Q}$ a task reports, candidate c is scored by its worst relative validation error on a query window $\mathcal{W}$, $S(c, \mathcal{W}) = \max_{q \in \mathcal{Q}} \mathcal{L}_q(\hat{\mathbf{y}}^{(c)}, \mathbf{y}; \mathcal{W}) / \mathcal{L}_q(\hat{\mathbf{y}}, \mathbf{y}; \mathcal{W})$, so a candidate must not degrade any reported metric to be preferred. Adding the dial multiplies the candidate pool sevenfold, which makes the minimizer of a single score optimistically biased, so TIMERAF scores each candidate on the worse of the two chronological halves of the query window, $S(c) = \max\{S(c, \mathcal{W}_1), S(c, \mathcal{W}_2)\}$: a correction that helps early and hurts late can no longer be selected. The minimizing family and its parameters are locked and applied once to test; ties go to identity. Because the identity candidate scores exactly one, the gate can always refuse to act.

Nothing in the rule privileges one retrieval object: β is chosen exactly as α , k , or a seasonal period is, by whichever candidate survives both halves of the query window. Section 4.2 reports where that leaves the dial and what a horizon-dependent prior on it would add.

Causality and finiteness. Predictions, residuals, corrections, and metrics must all be finite, and a non-finite value is a failed attempt rather than a result. Online statistics are indexed by target timestamp, so no correction can read a label that has not elapsed; as a check, perturbing every not-yet-realized target must leave the retrieval set, the online statistics, and the emitted correction identical. Appendix D reports these checks per setting.

4 EXPERIMENTS

We evaluate TIMERAF on long-term and short-term forecasting over 16 real-world benchmarks and 13 base forecasting models. Section 4.1 reports the main results and Section 4.2

Table 1: **Long-term forecasting.** Each entry averages 13 backbones and horizons {96, 192, 336, 720} over the *same* 364 base forecasts; ANALOG-KNN and RESIDUAL-KNN pin Equation 4 at $\beta = 1$ and $\beta = 0$. Lower is better; **best** and second best per column group.

Dataset	Uncorrected	Retrieve inputs / futures			Retrieve errors (ours)	
MSE / MAE	Base	ANALOG-KNN	RAFT	SARAF	RESIDUAL-KNN	TIMERAF
ETTh1	0.529 0.497	0.508 0.487	0.502 0.486	<u>0.501</u> <u>0.485</u>	0.519 0.494	0.481 0.471
ETTh2	0.597 0.515	<u>0.431</u> <u>0.438</u>	0.441 0.442	0.443 0.444	0.544 0.493	0.427 0.436
ETTm1	0.481 0.453	<u>0.442</u> <u>0.437</u>	0.455 0.441	0.455 0.441	0.447 0.440	0.428 0.431
ETTm2	0.385 0.394	<u>0.295</u> <u>0.344</u>	0.305 0.352	0.305 0.352	0.364 0.383	0.291 0.342
Weather	0.321 0.338	<u>0.286</u> <u>0.319</u>	0.292 0.322	0.293 0.323	0.289 0.324	0.272 0.310
Electricity	0.242 0.329	<u>0.207</u> <u>0.305</u>	0.213 0.308	0.212 0.308	0.216 0.310	0.205 0.301
Traffic	0.597 0.361	0.574 0.354	0.579 0.356	0.575 0.356	0.550 <u>0.343</u>	<u>0.556</u> 0.342
Average	0.450 0.413	<u>0.392</u> <u>0.383</u>	0.398 0.387	0.398 0.387	0.418 0.398	0.380 0.376

analyses which correction each deployment ends up using; full configurations are in Appendix C.

Datasets. For long-term forecasting we use seven standard benchmarks at horizons {96, 192, 336, 720}: the four ETT subsets, Weather, Electricity, and Traffic (Wang et al., 2024b). For short-term forecasting we use the PEMS traffic networks PEMS03/04/07/08 (Chen et al., 2001) at {6, 12, 24} and the EPF electricity markets NP, PJM, BE, FR, and DE (Lago et al., 2021) at horizon 24. One dataset-backbone-horizon combination is one *setting*, giving 585 in total: 364 long-term, 156 PEMS, and 65 EPF (Appendix Table 5).

Base models and setup. We correct 13 forecasters implemented in TSLib (Wang et al., 2024b) with their official configurations, L2 loss, and seed 2021: TIMEXER, TIMEMIXER, PATTN, ITRANSFORMER, TIMESNET, PATCHTST, DLINEAR, FRETTS, FEDFORMER, NON-STATIONARY TRANSFORMER, LIGHTTS, INFORMER, and AUTOFORMER (Wang et al., 2024c;a; Tan et al., 2024; Liu et al., 2024b; Wu et al., 2023; Nie et al., 2023; Zeng et al., 2023; Yi et al., 2024; Zhou et al., 2022; Liu et al., 2022; Zhang et al., 2022; Zhou et al., 2021; Wu et al., 2021). A backbone is trained once and never modified; validation predictions build memory, validation labels lock one correction per setting, and that correction is applied once to test, so test labels never enter selection (Appendix D). Every compared system additionally receives byte-identical base forecasts under one validation split, memory boundary, and scoring rule, with all grids fixed in advance (Appendix A), so accuracy differences can only come from what is retrieved. Long-term and EPF settings report MSE and MAE in normalized space and PEMS reports inverse-scaled MAE, RMSE, and MAPE; a *strict win* is a setting in which every metric it reports decreases. Because base errors span an order of magnitude, we report medians of paired gains alongside means.

Compared systems. Equation 4 lets us compare retrieval objects rather than implementations. ANALOG-KNN pins it at $\beta = 1$, injecting from the same neighbourhood and descriptor the endpoint-aligned futures that input-retrieval systems emit, and RESIDUAL-KNN pins it at $\beta = 0$. RAFT (Han et al., 2025) and SARAF (Zhou et al., 2026) contribute two published operators on the same fixed forecasts, while TIMERAF chooses α , β , and the horizon profile per setting on validation.

4.1 MAIN RESULTS

Long-term forecasting. In Table 1, TIMERAF attains the lowest average MSE and MAE, 0.380/0.376 against 0.450/0.413 uncorrected, and is best on both metrics for six of the seven datasets; only on Traffic is RESIDUAL-KNN lower on MSE. The two pinned objects disagree rather than one dominating: retrieving futures averages 0.392/0.383 against 0.418/0.398 for retrieving errors, and neither leads a dataset once the object is chosen per setting. That

Table 2: **Short-term forecasting.** Same protocol and columns on the PEMS networks (13 backbones, horizons $\{6, 12, 24\}$) and the five EPF markets (horizon 24); PEMS RMSE is in Appendix A. Lower is better; **best** and **second best** per column group.

Dataset	Uncorrected	Retrieve inputs / futures			Retrieve errors (ours)	
Metrics	Base	ANALOG-KNN	RAFT	SARAF	RESIDUAL-KNN	TIMERAF
<i>PEMS traffic flow, MAE / MAPE, inverse-scaled (156 settings)</i>						
PEMS03	20.781 0.208	<u>17.097</u> <u>0.168</u>	19.855 0.196	19.886 0.196	18.224 0.182	16.275 0.160
PEMS04	26.982 0.189	<u>22.541</u> <u>0.149</u>	25.911 0.179	25.972 0.180	23.664 0.161	21.467 0.140
PEMS07	30.087 0.139	<u>24.210</u> <u>0.104</u>	28.865 0.133	28.940 0.133	26.224 0.118	23.006 0.098
PEMS08	23.086 0.148	<u>18.069</u> <u>0.114</u>	22.005 0.139	22.049 0.139	19.654 0.127	16.878 0.108
Average	25.234 0.171	<u>20.479</u> <u>0.134</u>	24.159 0.162	24.212 0.162	21.942 0.147	19.407 0.126
<i>EPF electricity price, MSE / MAE (65 settings)</i>						
NP	0.314 0.328	0.295 0.311	0.294 0.311	<u>0.291</u> <u>0.309</u>	0.301 0.320	0.284 0.303
PJM	0.121 0.224	0.110 0.216	0.110 0.216	<u>0.109</u> 0.216	0.114 <u>0.216</u>	0.103 0.207
BE	0.453 0.305	0.451 0.296	0.447 0.298	0.448 0.300	<u>0.425</u> <u>0.294</u>	0.425 0.284
FR	0.465 0.258	0.469 <u>0.242</u>	0.455 0.248	0.457 0.246	<u>0.453</u> 0.244	0.450 0.225
DE	0.526 0.459	0.506 <u>0.449</u>	0.506 0.449	<u>0.505</u> 0.449	0.520 0.451	0.481 0.433
Average	0.376 0.315	0.366 <u>0.303</u>	0.362 0.304	<u>0.362</u> 0.304	0.363 0.305	0.349 0.290

choice is what recovers both, with median paired reductions of 4.17% MSE and 2.62% MAE against 2.27%/0.90% for ANALOG-KNN and 2.41%/1.20% for RESIDUAL-KNN.

Short-term forecasting. Table 2 shows a wider margin on shorter horizons, where TIMERAF is best on both reported metrics for all nine datasets. On PEMS it averages 19.407 MAE and 0.126 MAPE against 25.234/0.171 uncorrected and 20.479/0.134 for the strongest pinned object, with median paired reductions of 17.77% MAE, 15.95% RMSE, and 20.14% MAPE. On EPF it averages 0.349/0.290 against 0.376/0.315 and 0.362/0.303, reducing the median MSE by 4.57% and MAE by 5.44%. The published operators are close to inactive in this regime, with median PEMS reductions of 0.00% RMSE and 0.36%/0.31% MAPE, because multiscale input matching returns neighbourhoods dominated by the traffic profile the backbone already predicts.

Improvement is broad, not only on average. An average can hide a correction that helps some settings and hurts others, so we also count settings in which *every* reported metric decreases, which TIMERAF achieves in 541 of 585: 326/364 long-term, 156/156 PEMS, and 59/65 EPF, against 490 and 478 for the two pinned objects and 384 and 375 for SARAF and RAFT, which lose mostly by having nothing to contribute, returning a forecast identical to the base in 121 and 131 settings against 50 for RESIDUAL-KNN and 3 for TIMERAF. Compared setting by setting, TIMERAF improves every metric relative to ANALOG-KNN in 453 settings against 54 the other way, and relative to RESIDUAL-KNN in 457 against 63. On the five most accurate backbones the counts are 190/225 against 170, 157, 80, and 77.

Correcting 13 end-to-end trained forecasters. The comparison above holds base forecasts fixed to isolate the retrieval object. On the separate primary matrix, where each backbone is trained and corrected end to end, the error-only form of TIMERAF—Equation 4 at $\beta \equiv 0$ with single-window selection—improves every reported metric in 562/585 settings (96.1%): 355/364 long-term, 151/156 PEMS, and 56/65 EPF, with mean paired reductions of 8.5% MSE, 7.5% MAE, 10.6% RMSE, and 12.0% MAPE. Gains scale with how much error there is to remove, from 1.1% long-term MSE for TIMEXER to 41.3% for INFORMER, as an additive layer should, and no backbone is left behind (Appendix Table 17). The selection rule was fixed on a predetermined 123-setting development subset, and on the 462 held-out settings the strict-win rate is 437/462 (Appendix E). A locked correction also survives the passage of time: replaying TIMEMIXER at horizon 96 on a timeline beginning after the standard test split, with labels released only as their targets elapse, improves both metrics on all four ETT datasets and three seeds (Appendix H).



Figure 3: **The better retrieval object depends on how good the underlying forecaster already is.** RESIDUAL-KNN minus ANALOG-KNN median worst-metric gain (percentage points, symmetric-log axis) per backbone, ordered from most to least accurate by long-term base MSE.

Ensembles, AutoML, and foundation models. Because TIMERAf consumes only predictions and targets, the same interface corrects systems that are not single models. Appendix Table 34 reports 81/94 strict wins over greedy forward selection, a portfolio and a zero-shot ensemble, AUTOGLUON’s `high_quality` predictor, and CHRONOS-BOLT in fine-tuned and zero-shot form (Caruana et al., 2004; Feurer et al., 2022; Shchur et al., 2023; Ansari et al., 2024), from 12/16 to 16/16 per system, with the retrieval family selected in 70 settings and 65 of those strict wins: multi-component systems and an off-distribution foundation model still leave error structure their own machinery does not remove.

4.2 FURTHER ANALYSIS

Which object is better depends on how strong the forecaster is. Ordering the 13 backbones by long-term base MSE, the advantage of retrieving errors over retrieving futures falls as base error rises in both task families, with Spearman ρ of -0.94 and -0.77 and a sign change at TIMESNET (Figure 3). The mechanism is the drift term of Equation 1: a weak forecaster’s prediction holds little level, phase, or rhythm worth keeping, so replacing it with an analogous future is an improvement, whereas an accurate forecaster’s prediction already holds all of that and only its residual is worth transferring. INFORMER, whose long-term base MSE is three times the best backbone’s, is what pulls the ETTh2 and ETTm2 averages towards ANALOG-KNN.

The selected object follows the forecast horizon. Validation puts non-zero weight on the drift term in 435 of 585 settings, and where it does declines with the horizon: 50/52, 50/52, and 112/117 settings at horizons 6, 12, and 24, then 69/91, 65/91, 64/91, and 25/91 at 96 through 720. A neighbour’s trajectory stays usable for a few hours of traffic flow and is least usable a month ahead, where the weight survives in barely a quarter of settings. It is also rarely extreme, as 299 of those 435 settings use $\beta \in \{0.25, 0.5\}$ rather than the full overwrite input retrieval performs, so the selected operating point is one neither published object can express. Retrieval of either kind is selected in 449 of the 585 settings, taking all 156 PEMS and 63 of 65 EPF settings and 230 of 364 long-term ones, where a seasonal blend takes 75 and an overlap blend 46 (Appendix Figure 5).

What the two-fold score and the free dial each contribute. Scoring candidates on the worse of the two chronological validation halves instead of the whole window raises strict wins from 533 to 552 with the dial pinned shut. Opening the dial then trades 11 of those wins for magnitude everywhere: the median worst-metric gain rises from 9.73% to 15.69% on PEMS, from 1.77% to 3.56% on EPF, and from 3.64% to 4.17% in long-term MSE. Restricting $\beta > 0$ to horizons of at most 24 steps unless it clears the best $\beta = 0$ candidate by 0.05 recovers those 11 wins, reaching 555/585, but gives back the long-term magnitude (median MSE 3.64% again). We report the free dial, because the choice between per-setting accuracy and per-setting win rate is a deployment preference rather than a property of the method.

Where the correction layer fails. Of the 585 settings, 44 are not strict wins: 38 long-term and 6 EPF, none on PEMS, concentrated on ETTm1 (17), ETTh1 (8), ETTh2 (7),

and EPF-FR (6), and on the most accurate backbones, where the base trajectory is the thing worth protecting. Only 7 degrade every metric and 3 leave the forecast untouched; the other 34 trade one metric for another, the largest single loss being 6.9% MSE on an EPF-FR setting that gains 16.4% MAE. Gains can also be thin, with the worst metric improving by at most one point in 68 of the 541 strict wins, and selecting on a single validation window fails harder: 15 of its 52 non-strict settings degrade every metric.

5 RELATED WORK

Time-series forecasting. Classical models encode trend and seasonality explicitly (Anderson, 1976), while neural forecasters attack the same structure with frequency decomposition, patching, cross-variable attention, multiscale mixing, exogenous conditioning, linear baselines, or large-corpus pretraining (Zhou et al., 2022; Wu et al., 2023; Nie et al., 2023; Tan et al., 2024; Liu et al., 2024b; Wang et al., 2024a;c; Zeng et al., 2023; Zhang et al., 2022; Yi et al., 2024; Ansari et al., 2024). All improve the forecast by changing the forecaster; we show that even the strongest of them leaves a recurring, retrievable error behind.

Retrieval-augmented forecasting. Retrieval has supplied cross-entity context (Yang et al., 2022), guided a diffusion sampler (Liu et al., 2024a), populated LLM prompts (Yang et al., 2025), and conditioned foundation models (Tire et al., 2024). Zhang et al. (2024) train a retriever with channel prompting under the same acronym as our method; TS-RAG adds an encoder and adaptive mixer (Ning et al., 2025); RAFT targets recurring patterns model weights miss (Han et al., 2025); and CROSS-RAG, CRAFT, and SARAF sharpen relevance through cross-attention, spectral ranking, and stationarity-aware diversity (Lee et al., 2026; Kang et al., 2026; Zhou et al., 2026). All retrieve on the input and inject trajectories or latents into a learned prediction path. We change the retrieval object instead, and show it is one end of a dial their formulation fixes.

Forecast combination and post-hoc correction. Ensembles improve robustness by combining predictors (Kourentzes et al., 2014; Oliveira & Torgo, 2015; Mienye & Sun, 2022; Caruana et al., 2004; Feurer et al., 2022), AUTOGLUON automates selection, stacking, and weighting (Shchur et al., 2023), and TIMEFUSE learns sample-dependent fusion weights (Liu et al., 2025). These recombine forecasts; TIMERAf composes with them, asking whether the forecast a system produces has a recurring retrievable error—which, in 70 of 94 such settings, it does. Bias correction and residual boosting are classical special cases of our candidate set, and their presence makes the retrieval claim falsifiable.

6 CONCLUSION

We asked what a retrieval-augmented time-series forecaster should retrieve, and answered that it should retrieve its own errors—and that the futures its predecessors retrieve are the same correction with one term forced on. TIMERAf realizes that answer as a post-hoc layer over a chronological memory of a forecaster’s validation residuals, with shrinkage, a dial over the retrieval object, and a gate that can leave the forecast untouched. With all 585 base forecasts fixed it reaches 541 strict wins against 490 for pure error retrieval and 478 for the analogue control, and takes the best average in every task family.

Scope and limitations. Error memory pays off when a forecaster’s mistakes recur and borrowing a neighbour’s trajectory pays off when they do not; Figure 3 locates that crossover by base accuracy and Section 4.2 by horizon. Leaving the dial free costs 11 strict wins against pinning it shut, and a horizon-dependent prior on it recovers those wins while giving back most of the long-term magnitude, so which of the two to deploy is a choice we report rather than settle. The end-to-end and additional-system results use the error-only configuration, so what the dial adds is measured only with base forecasts fixed. The primary matrix uses one seed and the prequential study one backbone; descriptor, pooling, and shrinkage ablations remain open.

AI USE STATEMENT

Generative AI tools assisted with literature discovery, code, plotting, and manuscript editing. The authors verified all citations, analyses, experimental measurements, and final text.

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APPENDIX CONTENTS

The appendix follows the paper’s questions rather than the order in which experiments were run. Each entry below states what its evidence is for.

Section	Purpose
Appendix A, p. 12	Matched fixed-forecast retrieval controls and per-backbone values
Appendix B, p. 14	Datasets, models, training scope, and hardware
Appendix C, p. 15	Exact validation search space and selection rule
Appendix D, p. 17	Information boundaries and no-lookahead checks
Appendix E, p. 18	Policy attribution, failures, and hardware sensitivity
Appendix F, p. 18	Primary-matrix results by horizon, backbone, and cell
Appendix G, p. 31	Ensemble, AutoML, and foundation-model results
Appendix H, p. 32	Three-seed forward-transfer study on later observations

A RETRIEVAL COMPARISONS AND CONTROLS

The matched fixed-forecast comparison spans the same 585 backbone–dataset–horizon cells for every method. RAFT uses multiscale Pearson retrieval and softmax aggregation (Han et al., 2025); SARAF adds time alignment, stationarity-controlled relevance–diversity selection, and Gaussian aggregation (Zhou et al., 2026). These rows evaluate retrieval operators on common base forecasts, not the original learned forecasting or fusion layers.

ANALOG-KNN transfers a normalized historical future, whereas RESIDUAL-KNN transfers the corresponding error from the same base model; in the notation of Equation 4 they pin $\beta = 1$ and $\beta = 0$. The full TIMERAf row selects β , the horizon profile, and the alternative simple corrections on validation. All methods select parameters on the same chronological validation suffix; retrieval memory ends at least one horizon earlier, and prediction APIs receive no test labels. The TIMERAf row was produced by a later run than the comparison systems; its error-only configuration reproduces the earlier run’s selected candidate in all 585 cells and every strict-win classification, with a maximum relative metric difference of 1.6×10^{-5} from thread-level float reordering, so the rows remain paired cell by cell. Tables 3–4 give the per-backbone absolute errors and paired deltas behind main Tables 1 and 2, including the PEMS RMSE column omitted there. The accompanying machine-readable results retain all 585 per-cell records, so duplicating them as 90 multi-page tables is unnecessary.

Table 3: **Long-term fixed-forecast comparison.** Mean over all 7 datasets and 4 horizons for each base model. Δ = method – Base; lower is better. Bold marks the lowest error and corresponding delta.

System	TXER	TMIX	PATTN	iTRANS.	TNET	PTST	DLIN	FRETS	FEDF.	NonStat	LTS	INF.	AUTOF
<i>Long-term, MSE (all datasets and horizons)</i>													
Base	0.335	0.352	0.360	0.356	0.380	0.350	0.421	0.405	0.422	0.521	0.452	1.053	0.447
RAFT	0.334	0.350	0.358	0.353	0.373	0.348	0.400	0.390	0.405	0.455	0.417	0.576	0.416
Δ	-0.001	-0.002	-0.003	-0.003	-0.007	-0.002	-0.021	-0.015	-0.018	-0.066	-0.035	-0.476	-0.031
SARAF	0.334	0.349	0.357	0.353	0.372	0.347	0.399	0.390	0.404	0.455	0.416	0.581	0.416
Δ	-0.001	-0.003	-0.003	-0.003	-0.008	-0.003	-0.022	-0.015	-0.019	-0.066	-0.036	-0.472	-0.031
TIMERAF	0.331	0.346	0.352	0.345	0.363	0.343	0.375	0.376	0.382	0.421	0.397	0.511	0.397
Δ	-0.004	-0.006	-0.008	-0.011	-0.017	-0.007	-0.046	-0.029	-0.041	-0.100	-0.055	-0.542	-0.050
<i>Long-term, MAE (all datasets and horizons)</i>													
Base	0.341	0.349	0.356	0.356	0.363	0.354	0.406	0.393	0.409	0.448	0.429	0.732	0.428
RAFT	0.341	0.348	0.355	0.356	0.362	0.353	0.394	0.385	0.399	0.412	0.407	0.509	0.409
Δ	0.000	0.000	0.000	-0.001	-0.002	-0.001	-0.012	-0.008	-0.010	-0.036	-0.023	-0.223	-0.018
SARAF	0.341	0.348	0.355	0.355	0.361	0.353	0.394	0.385	0.399	0.412	0.407	0.513	0.410
Δ	0.000	0.000	0.000	-0.001	-0.002	-0.001	-0.012	-0.008	-0.010	-0.036	-0.023	-0.219	-0.018
TIMERAF	0.339	0.346	0.352	0.350	0.356	0.350	0.377	0.376	0.384	0.396	0.391	0.477	0.395
Δ	-0.001	-0.003	-0.003	-0.006	-0.007	-0.003	-0.028	-0.018	-0.025	-0.052	-0.038	-0.255	-0.033

Table 4: **PEMS and EPF fixed-forecast comparison.** Dataset-horizon mean by base model within each task. Δ = method – Base; lower is better. Bold marks the lowest error and corresponding delta.

System	TXER	TMIX	PATTN	iTRANS.	TNET	PTST	DLIN	FRETS	FEDF.	NonStat	LTS	INF.	AUTOF
<i>PEMS, MAE (all datasets and horizons)</i>													
Base	23.27	18.32	24.51	22.17	22.19	25.71	30.23	22.83	31.00	21.63	23.71	23.70	38.76
RAFT	23.27	18.32	24.51	22.17	21.80	25.72	29.78	22.83	27.25	21.00	23.16	22.47	31.80
Δ	0.00	0.00	0.00	0.00	-0.40	+0.01	-0.45	0.00	-3.76	-0.63	-0.55	-1.23	-6.96
SARAF	23.27	18.32	24.51	22.17	21.84	25.71	29.86	22.83	27.33	21.06	23.21	22.57	32.08
Δ	0.00	0.00	0.00	0.00	-0.36	0.00	-0.37	0.00	-3.67	-0.57	-0.50	-1.14	-6.68
TIMERAF	18.86	17.68	19.77	19.00	19.06	19.43	18.82	19.10	20.56	18.98	19.50	19.37	22.16
Δ	-4.41	-0.64	-4.74	-3.17	-3.14	-6.29	-11.41	-3.73	-10.44	-2.65	-4.21	-4.33	-16.60
<i>PEMS, RMSE (all datasets and horizons)</i>													
Base	35.00	28.85	38.26	34.99	34.81	39.22	44.86	35.56	44.85	34.12	35.73	37.61	55.18
RAFT	35.00	28.85	38.26	34.98	33.87	39.22	44.31	35.56	39.78	32.88	35.04	35.06	45.50
Δ	0.00	0.00	0.00	-0.01	-0.94	+0.01	-0.55	0.00	-5.07	-1.24	-0.69	-2.55	-9.68
SARAF	35.00	28.85	38.26	34.98	33.97	39.22	44.38	35.56	39.88	32.99	35.09	35.30	45.85
Δ	0.00	0.00	0.00	-0.01	-0.85	0.00	-0.48	0.00	-4.97	-1.13	-0.64	-2.31	-9.32
TIMERAF	29.92	28.13	31.56	30.41	30.12	31.05	30.04	30.53	31.79	30.06	30.55	30.70	33.59
Δ	-5.08	-0.72	-6.70	-4.57	-4.70	-8.16	-14.81	-5.03	-13.06	-4.06	-5.18	-6.91	-21.59
<i>PEMS, MAPE (all datasets and horizons)</i>													
Base	0.157	0.121	0.154	0.141	0.146	0.165	0.219	0.147	0.228	0.142	0.162	0.157	0.284
RAFT	0.157	0.121	0.154	0.141	0.143	0.165	0.212	0.147	0.196	0.137	0.155	0.148	0.225
Δ	0.000	0.000	0.000	0.000	-0.003	0.000	-0.007	0.000	-0.032	-0.005	-0.006	-0.009	-0.059
SARAF	0.157	0.121	0.154	0.141	0.143	0.165	0.214	0.147	0.197	0.137	0.156	0.148	0.227
Δ	0.000	0.000	0.000	0.000	-0.003	0.000	-0.005	0.000	-0.031	-0.005	-0.006	-0.008	-0.057
TIMERAF	0.121	0.115	0.126	0.120	0.124	0.124	0.121	0.123	0.137	0.122	0.129	0.125	0.153
Δ	-0.036	-0.006	-0.028	-0.021	-0.023	-0.041	-0.098	-0.024	-0.090	-0.020	-0.033	-0.031	-0.131
<i>EPF, MSE (all datasets and horizons)</i>													
Base	0.317	0.339	0.343	0.331	0.327	0.341	0.358	0.362	0.397	0.544	0.363	0.398	0.469
RAFT	0.318	0.337	0.338	0.327	0.322	0.340	0.356	0.357	0.371	0.487	0.359	0.379	0.419
Δ	0.000	-0.002	-0.005	-0.003	-0.005	-0.001	-0.001	-0.005	-0.027	-0.057	-0.004	-0.019	-0.050
SARAF	0.318	0.337	0.336	0.331	0.321	0.339	0.355	0.355	0.369	0.491	0.357	0.377	0.420
Δ	+0.001	-0.002	-0.007	+0.001	-0.006	-0.002	-0.003	-0.007	-0.029	-0.053	-0.006	-0.021	-0.049
TIMERAF	0.314	0.332	0.328	0.316	0.313	0.335	0.353	0.347	0.358	0.434	0.347	0.361	0.396
Δ	-0.004	-0.007	-0.015	-0.014	-0.014	-0.006	-0.005	-0.015	-0.039	-0.110	-0.015	-0.037	-0.073
<i>EPF, MAE (all datasets and horizons)</i>													
Base	0.269	0.294	0.297	0.286	0.286	0.287	0.310	0.312	0.344	0.412	0.305	0.328	0.364
RAFT	0.268	0.291	0.293	0.284	0.282	0.286	0.307	0.306	0.317	0.372	0.302	0.314	0.335
Δ	-0.001	-0.003	-0.004	-0.003	-0.003	-0.001	-0.002	-0.006	-0.027	-0.040	-0.003	-0.014	-0.029
SARAF	0.268	0.291	0.292	0.287	0.282	0.285	0.306	0.305	0.317	0.368	0.302	0.315	0.336
Δ	-0.001	-0.003	-0.005	0.000	-0.004	-0.002	-0.004	-0.007	-0.027	-0.043	-0.004	-0.013	-0.028
TIMERAF	0.263	0.283	0.283	0.275	0.276	0.280	0.289	0.295	0.298	0.328	0.292	0.301	0.311
Δ	-0.005	-0.011	-0.014	-0.011	-0.010	-0.007	-0.021	-0.017	-0.046	-0.084	-0.013	-0.026	-0.053

B DATASET AND MODEL SCOPE

Table 5: Full benchmark scope. “Cells” includes all 13 base models.

Family	Dataset	Variates	Input	Horizons	Cells	Frequency
Long-term	ETTh1	7	96	96,192,336,720	52	1 hour
	ETTh2	7	96	96,192,336,720	52	1 hour
	ETTm1	7	96	96,192,336,720	52	15 min
	ETTm2	7	96	96,192,336,720	52	15 min
	Weather	21	96	96,192,336,720	52	10 min
	Electricity	321	96	96,192,336,720	52	1 hour
	Traffic	862	96	96,192,336,720	52	1 hour
PEMS	PEMS03	358	96	6,12,24	39	5 min
	PEMS04	307	96	6,12,24	39	5 min
	PEMS07	883	96	6,12,24	39	5 min
	PEMS08	170	96	6,12,24	39	5 min
EPF	NP	1	168	24	13	1 hour
	PJM	1	168	24	13	1 hour
	BE	1	168	24	13	1 hour
	FR	1	168	24	13	1 hour
	DE	1	168	24	13	1 hour
Total					585	

The model list is TIMEXER, TIMEMIXER, PATTN, iTRANSFORMER, TIMESNET, PATCHTST, DLINEAR, FRETTS, FEDFORMER, NON-STATIONARY TRANSFORMER, LIGHTTS, INFORMER, and AUTOFORMER. Baseline training follows each paper configuration in TSLib. Shared defaults are Adam optimization, MSE training loss, at most 10 epochs, and validation-loss early stopping unless the referenced configuration overrides them. NON-STATIONARY TRANSFORMER uses a learning rate of 10^{-4} on Electricity horizons 192 and 336; all other values follow the reference configuration.

The primary matrix trains all 585 model–dataset–horizon cells under these configurations with seed 2021, PyTorch 2.8.0, and CUDA 12.9 on one `m1.p4de.24xlarge` host with eight NVIDIA A100 GPUs. Independent cells use one GPU each. A secondary execution on an `m1.g5.12xlarge` host uses 55 compatible released checkpoints and trains the remaining 530 cells on four NVIDIA A10G GPUs. It is retained only as a cross-hardware sensitivity check and is not averaged into the primary tables because the two executions are not numerically identical. The additional-system study uses AUTOGLUON-TimeSeries 1.4.0, Python 3.12, PyTorch 2.7.1, and a three-hour fit budget for each AUTOGLUON cell.

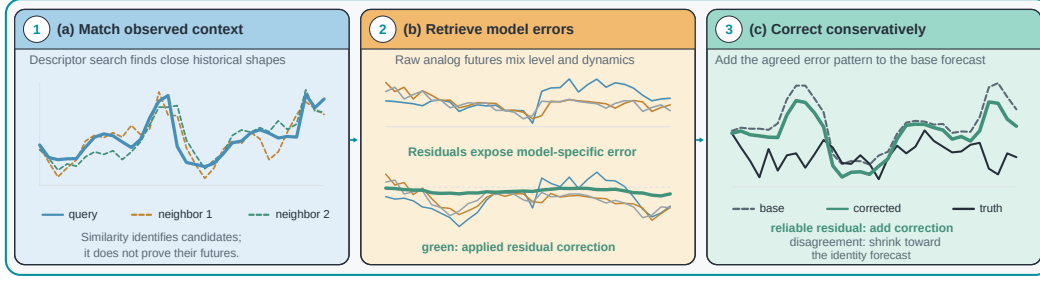


Figure 4: **Residuals isolate model-specific error on real data.** An ETTh1 MULL example from a trained TIMEMIXER. (a) Descriptor search returns historical windows with closely matching shapes. (b) Their raw futures sit at different levels and mix dynamics the base forecast already captures, whereas their residuals agree on a common error pattern. (c) Adding the shrunk residual moves the forecast toward the truth without disturbing its level. Panels (b)–(c) show a 36-step excerpt in the model’s standardized space.

Algorithm 1: TIMERAF calibration and forecast-time correction

Input: trained forecaster f ; validation stream $\{(\mathbf{x}_i, \mathbf{y}_i)\}$; candidate families and grids; metric set $\mathcal{Q}$.

Calibration. (1) For every validation origin i , evaluate $\hat{\mathbf{y}}_i = f(\mathbf{x}_i)$ and, once $\mathbf{y}_i$ has fully elapsed, store $(\phi_i, \hat{\mathbf{y}}_i, \mathbf{r}_i = \mathbf{y}_i - \hat{\mathbf{y}}_i)$ in $\mathcal{M}$. (2) Split $\mathcal{M}$ chronologically into a memory prefix and a query suffix. (3) For each candidate c , form $\tilde{\mathbf{y}}^{(c)}$ on the query suffix using only the memory prefix and score it by $S(c)$. (4) Lock the minimizing family and parameters c^* ; ties go to identity.

Forecasting at origin t . (5) Evaluate $\hat{\mathbf{y}}_t = f(\mathbf{x}_t)$ and build ϕ_t from $\mathbf{x}_t$ and $\hat{\mathbf{y}}_t$. (6) If c^* is retrieval, take the top- k items of $\mathcal{M}_t$ with $i + H < t$, aggregate by Equation 2 and shrink by Equation 3; otherwise evaluate the locked simpler correction. (7) Return $\tilde{\mathbf{y}}_t = \hat{\mathbf{y}}_t + c_t$, leaving f unchanged.

C EXACT BROAD SEARCH SPACE

The ETTh1 example in Figure 4 was screened deterministically for visualization only; the screening criteria and score weights are recorded in the released case artifact and never enter method selection.

Table 6: Validation-only candidate space for the broad selector.

Family	Candidate values
Identity	No parameters
Residual descriptor	Tail $\ell \in \{\min(48, L), L\}$; pooled steps $\min(12, \ell)$; differences on; base forecast on/off; calendar on except PEMS; at most 16 pooled channels
Residual aggregation	$(k, \tau, \lambda) \in \{(8, 0.1, 1), (16, 0.3, 4)\}$; $\alpha \in \{.05, .1, .2, .3, .5\}$
Seasonal blend	Long-term periods $\{4, 8, 12, 24, 48, 96, 168\}$, PEMS periods $\{12, 24, 48, 96\}$, and EPF periods $\{24, 48, 168\}$, each clipped at L and deduplicated; $\alpha \in \{.02, .05, .1, .2, .3\}$
Overlap blend	$(a, \gamma) \in \{(8, .75), (24, .9)\}$; $\alpha \in \{.1, .25, .5\}$
Static residual bias	$\alpha \in \{.1, .25, .5, 1\}$
Causal rolling bias	Window $W \in \{2H, 8H, 32H\}$; horizon block $b \in \{0, \min(24, H)\}$; $\alpha \in \{.1, .25, .5, 1\}$

Temporal-candidate parameters. For overlap blend, a is the maximum age in forecast origins and γ is the per-origin exponential decay. For causal rolling bias, W is the target-time lookahead. Setting $b = 0$ pools all forecast leads; otherwise, contiguous groups of b leads

Table 7: Information available to each correction candidate at forecast origin t . A label is eligible only after its target timestamp has passed.

Candidate	Information used
Identity	Current model prediction only; no labels
Seasonal blend	Observed context through t ; no labels
Overlap blend	Predictions issued at earlier origins for the same target timestamps; no labels
Static residual bias	Validation predictions and validation labels
Historical residual re-retrieval	Validation contexts, predictions, and labels
Causal rolling bias	Validation evidence and test labels with target time $u < t$
ETTM post-test re-retrieval	Validation evidence and completed official-test residuals; no post-test label with target time $u \geq t$
ETTh online stack	Training data for NLinear, earlier predictions of the same targets, and delayed labels with target time $u < t$

receive separate bias estimates. The grouping changes which completed errors are averaged, not when a label becomes eligible.

Chronological validation partition. Let validation contain n forecast windows. The query suffix begins at $\max(\lfloor n/2 \rfloor, H + 8)$, adjusted to stay below n . The residual memory ends at least H origins earlier, so every memory future is complete. The final test retriever is fit on the complete validation split because that split precedes test.

Selection tie breaking. Candidates minimize the worst-metric validation ratio $S(c)$ of Section 3.3. Identity wins an exact score tie. When the best score is at least 0.982, only candidates improving all validation metrics are eligible for family-prior selection. Within a family, the lowest score wins; the family order is overlap, seasonal, historical residual, static residual bias, and causal rolling bias.

Internal policy holdout. The supplementary material lists the fixed 123 development and 462 holdout cell IDs. Membership uses the first 123 completed cells ordered by `started_unix` and does not use any metric value. The boundary is nevertheless an opportunistic execution-order split rather than a stratified dataset, model, or difficulty holdout. Its scope and balance diagnostics are reported in Appendix E.

D NO-LOOKAHEAD PROTOCOL

Every forecast and residual is associated with both an origin and its target timestamps. Static validation memory is legal for test because all validation targets precede test. Within online evaluation, a contribution for target timestamp u becomes visible only for query origin $t > u$. Prefix arrays are queried at t , never $t + H$. Rolling-bias statistics write each residual component at its own target timestamp and query the strict prefix $u < t$; horizon blocks group forecast leads but cannot expose a partially realized future.

The evaluation enforces four conditions:

1. each retrieved residual’s final target is earlier than the query origin;
2. online statistics and weights contain only target-time evidence $u < t$, including when forecast windows overlap;
3. the PEMS timeline includes the fresh-context gap and 12-step test stride; and
4. post-test ETT memory may contain completed validation and official-test residuals but no current or future post-test target.

Table 7 states the corresponding information set for every candidate. Historical retrieval and static bias use validation labels; seasonal and overlap corrections use no labels; causal rolling bias may incorporate test-period labels only after their target timestamps have passed. Thus the broad selector contains both static and delayed-label prequential candidates.

As a direct no-lookahead check, we replace every label at timestamps $\geq t$ and require the correction at t to remain unchanged.

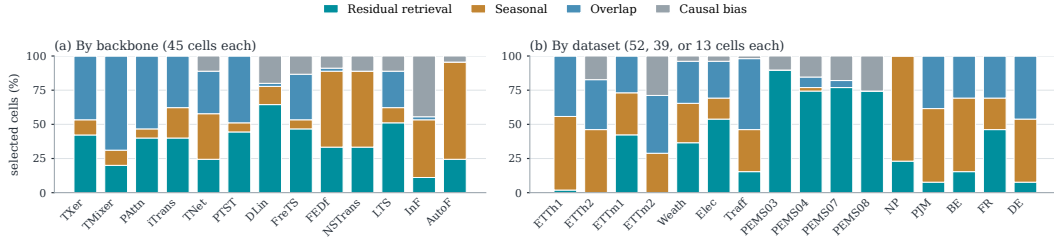


Figure 5: **Validation selects different corrections for different forecasters and regimes.** Share of the 585 primary settings in which each correction family was selected, grouped by backbone (a) and by dataset (b). Error retrieval is chosen for 29/45 DLINEAR settings but 5/45 INFORMER settings, and for 35/39 PEMS03 settings but 0/52 ETTh2 settings. A fixed retrieval-always policy would be wrong in a large fraction of deployments.

E AGGREGATE RESULTS

Table 8: Primary A100 outcomes stratified by the correction family selected on validation. This is descriptive attribution, not a portfolio-minus-retrieval ablation.

Selected family	Cells	Strict wins
Historical residual retrieval	214	207
Seasonal blend	156	151
Overlap blend	159	148
Causal rolling bias	56	56
Identity / static residual bias	0	0
Non-retrieval total	371	355

Table 9: A100 configurations for the 214 cells that select historical residual retrieval. Counts describe validation selections, not component ablations. The search couples (k, τ, λ) into two settings.

Quantity	Selected value	Cells
Correction scale α	.05 / .1 / .2 / .3 / .5	0 / 1 / 6 / 29 / 178
Aggregation (k, τ, λ)	(8, .1, 1) / (16, .3, 4)	151 / 63
Descriptor tail length	48 / 96 / 168	57 / 148 / 9
Base-forecast features	Excluded / included	61 / 153

All 214 selected retrieval configurations use first differences, 12 pooled time steps, and a maximum of 16 channel groups; these fixed choices are not ablated. Calendar features are present in 91 configurations and absent in 123, the latter corresponding to PEMS where calendar marks are unavailable.

Selected methods agree in 569/585 cells, selected parameters in 529/585, and strict-win classifications in 581/585. A100 remains the primary result; A10G is a same-seed sensitivity check.

F FULL BENCHMARK DECOMPOSITIONS

The main tables average matched cells by dataset while accounting for the fact that TIMERAf corrects each backbone independently. Table 13 and Tables 14–16 expose every dataset–horizon aggregate; each row contains exactly 13 paired backbone cells. EPF has only horizon 24; its complete per-model view appears in the full cell tables below. Table 17 instead aggregates by backbone and shows strict wins together with paired relative reductions. The remaining tables in this section report every one of the 585 primary cells. Each

Table 10: A100 correction-family selections by dataset. HR is historical residual retrieval, SB seasonal blend, OB overlap blend, and CB causal rolling bias. Identity and static bias have zero selections on every dataset.

Dataset	Cells	HR	SB	OB	CB	Dataset	Cells	HR	SB	OB	CB
ETTh1	52	1	28	23	0	ETTh2	52	0	24	19	9
ETTm1	52	22	16	14	0	ETTm2	52	0	15	22	15
Weather	52	19	15	16	2	Electricity	52	28	8	14	2
Traffic	52	8	16	27	1	PEMS03	39	35	0	0	4
PEMS04	39	29	1	3	6	PEMS07	39	30	0	2	7
PEMS08	39	29	0	0	10	NP	13	3	10	0	0
PJM	13	1	7	5	0	BE	13	2	7	4	0
FR	13	6	3	4	0	DE	13	1	6	6	0

Table 11: The 23 A100 cells that do not strictly improve, grouped by dataset. “Worst degradation” is the largest percentage increase among reported test metrics in that group. HR, SB, OB, and CB are defined in Table 10.

Dataset	Cells	HR	SB	OB	CB	Worst degradation
Weather	7	6	0	1	0	2.45%
DE	7	0	2	5	0	2.20%
PEMS04	3	0	0	3	0	36.50%
PEMS07	2	0	0	2	0	9.63%
ETTh2	1	0	1	0	0	1.12%
Electricity	1	1	0	0	0	0.33%
NP	1	0	1	0	0	2.64%
BE	1	0	1	0	0	1.53%
Total	23	7	5	11	0	36.50%

entry gives the absolute error of the named backbone followed by the error after applying TIMERAF; these tables do not average over backbones or horizons.

Table 12: A100–A10G discrepancy distribution over all 1,326 reported metric values. Relative difference is absolute difference divided by the larger absolute value. The pre-registered tolerance is $10^{-5} + 10^{-4} \max(|a|, |b|)$.

Quantity	Baseline	Corrected
Within tolerance	472/1326	469/1326
Median relative difference	0.09%	0.06%
90th percentile	2.95%	2.36%
95th percentile	5.80%	5.47%
Maximum	32.79%	28.13%

Table 13: Full long-term results by prediction length. Absolute metrics average the same 13 paired backbones; Δ is the mean cell-wise relative reduction (%).

Dataset	Horizon	Base MSE	+TiMERAF MSE	Δ MSE	Base MAE	+TiMERAF MAE	Δ MAE	Wins
ETTh1	96	0.462	0.412	7.8	0.454	0.421	6.0	13/13
ETTh1	192	0.499	0.457	5.7	0.478	0.451	4.5	13/13
ETTh1	336	0.565	0.516	6.4	0.513	0.484	4.7	13/13
ETTh1	720	0.607	0.544	7.3	0.553	0.522	4.5	13/13
ETTh2	96	0.452	0.322	12.8	0.437	0.368	9.8	13/13
ETTh2	192	0.574	0.401	13.4	0.499	0.416	10.0	12/13
ETTh2	336	0.634	0.446	14.9	0.540	0.450	10.7	13/13
ETTh2	720	0.727	0.569	11.9	0.584	0.525	7.1	13/13
ETTm1	96	0.411	0.355	11.1	0.415	0.386	6.0	13/13
ETTm1	192	0.455	0.403	8.8	0.437	0.409	5.5	13/13
ETTm1	336	0.502	0.442	8.7	0.463	0.435	4.9	13/13
ETTm1	720	0.571	0.502	9.4	0.501	0.471	4.8	13/13
ETTm2	96	0.241	0.193	10.1	0.310	0.279	6.8	13/13
ETTm2	192	0.315	0.255	9.7	0.358	0.319	7.4	13/13
ETTm2	336	0.417	0.317	13.4	0.417	0.357	9.8	13/13
ETTm2	720	0.566	0.404	14.4	0.492	0.408	10.9	13/13
Weather	96	0.223	0.194	8.9	0.271	0.250	5.6	13/13
Weather	192	0.271	0.243	6.8	0.309	0.291	4.1	9/13
Weather	336	0.348	0.287	8.4	0.356	0.320	5.9	10/13
Weather	720	0.435	0.358	7.6	0.410	0.367	5.7	13/13
Electricity	96	0.243	0.197	9.7	0.326	0.294	6.4	13/13
Electricity	192	0.204	0.191	5.3	0.300	0.288	3.7	12/13
Electricity	336	0.219	0.205	5.8	0.316	0.302	4.1	13/13
Electricity	720	0.307	0.261	8.3	0.379	0.347	6.0	13/13
Traffic	96	0.571	0.525	7.4	0.348	0.327	5.7	13/13
Traffic	192	0.579	0.537	6.4	0.353	0.332	5.5	13/13
Traffic	336	0.600	0.564	5.2	0.362	0.343	4.6	13/13
Traffic	720	0.642	0.608	4.5	0.382	0.366	3.8	13/13

Table 14: Full PEMS MAE results by prediction length. Absolute metrics average the same 13 paired backbones; Δ is the mean cell-wise relative reduction (%). Strict wins require all three PEMS metrics and therefore repeat across these tables.

Dataset	Horizon	Base MAE	+TIMERAF MAE	Δ MAE	Strict wins
PEMS03	6	18.504	16.378	10.2	13/13
PEMS03	12	20.037	17.545	11.6	13/13
PEMS03	24	24.055	20.647	13.2	13/13
PEMS04	6	23.807	21.872	7.4	12/13
PEMS04	12	26.688	23.728	9.5	12/13
PEMS04	24	31.475	26.896	12.2	12/13
PEMS07	6	26.781	24.330	8.8	12/13
PEMS07	12	29.145	25.604	11.3	12/13
PEMS07	24	34.510	29.108	14.6	13/13
PEMS08	6	20.024	18.071	9.6	13/13
PEMS08	12	22.363	19.316	12.6	13/13
PEMS08	24	27.167	22.449	15.9	13/13

Table 15: Full PEMS RMSE results by prediction length. Absolute metrics average the same 13 paired backbones; Δ is the mean cell-wise relative reduction (%). Strict wins require all three PEMS metrics and therefore repeat across these tables.

Dataset	Horizon	Base RMSE	+TIMERAF RMSE	Δ RMSE	Strict wins
PEMS03	6	28.874	25.601	10.2	13/13
PEMS03	12	31.319	27.566	11.3	13/13
PEMS03	24	37.664	32.623	12.4	13/13
PEMS04	6	36.313	33.846	6.3	12/13
PEMS04	12	40.393	36.414	8.7	12/13
PEMS04	24	47.375	40.790	12.0	12/13
PEMS07	6	40.478	37.340	7.5	12/13
PEMS07	12	44.220	39.634	9.7	12/13
PEMS07	24	51.621	44.486	12.9	13/13
PEMS08	6	30.493	27.795	8.8	13/13
PEMS08	12	33.811	29.497	11.9	13/13
PEMS08	24	41.074	34.084	15.6	13/13

Table 16: Full PEMS MAPE results by prediction length. Absolute metrics average the same 13 paired backbones; Δ is the mean cell-wise relative reduction (%). Strict wins require all three PEMS metrics and therefore repeat across these tables.

Dataset	Horizon	Base MAPE	+TIMERAF MAPE	Δ MAPE	Strict wins
PEMS03	6	0.187	0.164	10.5	13/13
PEMS03	12	0.197	0.174	11.1	13/13
PEMS03	24	0.238	0.207	12.4	13/13
PEMS04	6	0.166	0.147	10.2	12/13
PEMS04	12	0.189	0.165	10.3	12/13
PEMS04	24	0.221	0.188	11.8	12/13
PEMS07	6	0.122	0.107	11.8	12/13
PEMS07	12	0.132	0.114	12.8	12/13
PEMS07	24	0.165	0.135	16.2	13/13
PEMS08	6	0.125	0.111	10.2	13/13
PEMS08	12	0.141	0.122	12.7	13/13
PEMS08	24	0.174	0.147	14.5	13/13

Table 17: Backbone-level paired results. Entries are strict wins and mean cell-wise relative reductions (%); positive values favor TIMERAF. This view prevents weak backbones or large absolute errors from dominating the conclusion.

Backbone	Long-term (28 cells)			PEMS (12 cells)				EPF (5 cells)		
	Wins	MSE	MAE	Wins	MAE	RMSE	MAPE	Wins	MSE	MAE
TIMEXER	28/28	1.1	0.9	12/12	14.1	11.9	15.9	3/5	0.3	0.4
TIME MIXER	27/28	1.3	1.2	7/12	-2.4	-0.8	-5.5	4/5	3.2	2.5
PATTN	27/28	1.7	1.0	12/12	13.0	12.5	11.8	4/5	1.5	1.8
ITRANSFORMER	26/28	2.9	2.0	12/12	10.3	9.8	10.6	4/5	1.1	0.9
TIMESNET	28/28	2.9	2.0	12/12	5.8	4.8	7.1	4/5	4.8	3.2
PATCHTST	27/28	1.8	1.2	12/12	16.8	15.2	16.9	5/5	1.9	1.9
DLINEAR	27/28	10.1	7.6	12/12	24.7	23.4	27.4	5/5	4.7	4.7
FRETS	25/28	6.3	4.6	12/12	11.9	11.0	11.7	4/5	3.2	3.2
FEDFORMER	28/28	8.5	6.1	12/12	14.0	11.4	17.3	5/5	11.9	9.3
NON-STAT. TRANSFORMER	28/28	16.4	10.0	12/12	4.9	3.9	6.5	4/5	15.0	15.3
LIGHTTS	28/28	10.3	7.8	12/12	9.4	8.2	9.0	4/5	1.6	2.2
INFORMER	28/28	41.3	28.7	12/12	8.4	9.7	10.2	5/5	9.7	6.1
AUTOFORMER	28/28	11.6	8.0	12/12	17.4	17.0	17.4	5/5	17.5	10.7

Table 18: Complete long-term results on ETTh1. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	96		192		336		720	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TIMEKER	Base	0.3818	0.4029	0.4285	0.4355	0.4672	0.4494	0.4988	0.4926
	+TIMERAF	0.3739	0.3932	0.4267	0.4342	0.4622	0.4457	0.4893	0.4871
TIMEMIXER	Base	0.3784	0.3981	0.4410	0.4301	0.5003	0.4600	0.4787	0.4724
	+TIMERAF	0.3782	0.3895	0.4378	0.4278	0.4959	0.4568	0.4696	0.4680
PATTN	Base	0.3901	0.4048	0.4371	0.4392	0.4977	0.4703	0.5326	0.4995
	+TIMERAF	0.3828	0.3961	0.4346	0.4374	0.4915	0.4662	0.5195	0.4927
iTRANSFORMER	Base	0.4469	0.4439	0.5187	0.4833	0.5538	0.5018	0.5629	0.5251
	+TIMERAF	0.4054	0.4139	0.4679	0.4506	0.5107	0.4727	0.5124	0.4930
TIMESNET	Base	0.3890	0.4120	0.4395	0.4418	0.4941	0.4709	0.5178	0.4944
	+TIMERAF	0.3750	0.3933	0.4314	0.4258	0.4798	0.4519	0.5136	0.4928
PATCHTST	Base	0.3793	0.3996	0.4274	0.4325	0.4783	0.4643	0.5220	0.5063
	+TIMERAF	0.3727	0.3916	0.4266	0.4318	0.4736	0.4613	0.5121	0.5016
DLINER	Base	0.3962	0.4108	0.4450	0.4404	0.4874	0.4654	0.5126	0.5104
	+TIMERAF	0.3861	0.3990	0.4441	0.4394	0.4776	0.4608	0.5114	0.5099
FRETS	Base	0.3999	0.4124	0.4512	0.4426	0.5086	0.4814	0.5627	0.5398
	+TIMERAF	0.3928	0.4063	0.4436	0.4364	0.5000	0.4748	0.5497	0.5324
FEDFORMER	Base	0.3770	0.4184	0.4200	0.4441	0.4584	0.4666	0.5021	0.5032
	+TIMERAF	0.3643	0.3945	0.4133	0.4239	0.4526	0.4515	0.4974	0.5008
NON-STAT. TRANSFORMER	Base	0.5752	0.4933	0.4762	0.4783	0.7532	0.6289	0.8274	0.6659
	+TIMERAF	0.4516	0.4353	0.4336	0.4411	0.6365	0.5645	0.6365	0.5716
LIGHTTS	Base	0.4352	0.4436	0.4936	0.4782	0.5493	0.5103	0.6117	0.5688
	+TIMERAF	0.4133	0.4280	0.4653	0.4592	0.5169	0.4900	0.5748	0.5492
INFORMER	Base	0.9590	0.7778	1.0199	0.7930	1.0310	0.7809	1.2219	0.8817
	+TIMERAF	0.6328	0.6056	0.6812	0.6191	0.7092	0.6221	0.8044	0.6969
AUTOFORMER	Base	0.4993	0.4807	0.4868	0.4756	0.5648	0.5242	0.5426	0.5231
	+TIMERAF	0.4262	0.4328	0.4408	0.4400	0.5024	0.4801	0.4850	0.4849

Table 19: Complete long-term results on ETTh2. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	96		192		336		720	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TIMEKER	Base	0.2854	0.3376	0.3627	0.3892	0.4113	0.4214	0.4485	0.4534
	+TIMERAF	0.2842	0.3365	0.3606	0.3875	0.4076	0.4192	0.4481	0.4531
TIMEMIXER	Base	0.2878	0.3395	0.3778	0.3957	0.4247	0.4298	0.4681	0.4683
	+TIMERAF	0.2875	0.3391	0.3820	0.3917	0.4163	0.4246	0.4671	0.4674
PATTN	Base	0.2988	0.3453	0.3826	0.3954	0.4232	0.4305	0.4279	0.4438
	+TIMERAF	0.2941	0.3408	0.3782	0.3917	0.4182	0.4269	0.4233	0.4407
iTRANSFORMER	Base	0.3004	0.3496	0.3818	0.3998	0.4236	0.4324	0.4264	0.4451
	+TIMERAF	0.2997	0.3435	0.3787	0.3969	0.4202	0.4298	0.4231	0.4426
TIMESNET	Base	0.3370	0.3709	0.4046	0.4146	0.4520	0.4503	0.4337	0.4477
	+TIMERAF	0.3140	0.3536	0.3896	0.4015	0.4340	0.4357	0.4305	0.4455
PATCHTST	Base	0.2921	0.3452	0.3814	0.4044	0.4185	0.4349	0.4332	0.4524
	+TIMERAF	0.2890	0.3421	0.3752	0.4006	0.4103	0.4306	0.4298	0.4502
DLINER	Base	0.3414	0.3953	0.4819	0.4792	0.5929	0.5422	0.8403	0.6611
	+TIMERAF	0.2996	0.3483	0.3584	0.3902	0.4123	0.4255	0.6150	0.5612
FRETS	Base	0.3415	0.3967	0.4396	0.4492	0.5375	0.5094	0.7956	0.6436
	+TIMERAF	0.3013	0.3538	0.4045	0.4207	0.4720	0.4707	0.5933	0.5511
FEDFORMER	Base	0.3480	0.3906	0.4245	0.4371	0.4566	0.4676	0.4758	0.4858
	+TIMERAF	0.3190	0.3653	0.3994	0.4133	0.4388	0.4481	0.4502	0.4638
NON-STAT. TRANSFORMER	Base	0.4594	0.4509	0.6137	0.5253	0.7591	0.5913	0.5136	0.4918
	+TIMERAF	0.3688	0.3963	0.4933	0.4618	0.5975	0.5148	0.4538	0.4586
LIGHTTS	Base	0.4353	0.4730	0.5613	0.5382	0.6829	0.5985	0.9366	0.7145
	+TIMERAF	0.3615	0.4144	0.4604	0.4708	0.4795	0.4857	0.6603	0.5912
INFORMER	Base	1.7643	1.0737	2.2024	1.2022	2.1738	1.2303	2.7496	1.3864
	+TIMERAF	0.4310	0.4645	0.4212	0.4602	0.4386	0.4705	1.5481	1.0228
AUTOFORMER	Base	0.3817	0.4150	0.4512	0.4530	0.4833	0.4830	0.5035	0.5025
	+TIMERAF	0.3363	0.3848	0.4106	0.4257	0.4567	0.4620	0.4599	0.4732

Table 20: Complete long-term results on ETTm1. Base and +TimeRAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	96		192		336		720	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TimeXER	Base	0.3178	0.3563	0.3616	0.3829	0.3951	0.4067	0.4524	0.4413
	+TimeRAF	0.3072	0.3514	0.3583	0.3808	0.3872	0.4020	0.4470	0.4401
TimeMixer	Base	0.3184	0.3581	0.3654	0.3868	0.3894	0.4027	0.4697	0.4528
	+TimeRAF	0.3040	0.3515	0.3560	0.3805	0.3812	0.3974	0.4515	0.4453
PATtn	Base	0.3364	0.3648	0.3762	0.3832	0.4074	0.4036	0.4674	0.4376
	+TimeRAF	0.3201	0.3583	0.3737	0.3808	0.4048	0.4012	0.4533	0.4357
iTransformer	Base	0.3557	0.3813	0.3988	0.4020	0.4330	0.4242	0.5005	0.4615
	+TimeRAF	0.3379	0.3733	0.3973	0.4007	0.4315	0.4231	0.4800	0.4582
TimesNet	Base	0.3339	0.3754	0.4081	0.4137	0.4150	0.4223	0.5101	0.4722
	+TimeRAF	0.3180	0.3602	0.3771	0.3878	0.4134	0.4212	0.5034	0.4704
PATCHTST	Base	0.3268	0.3664	0.3700	0.3903	0.3982	0.4084	0.4577	0.4444
	+TimeRAF	0.3131	0.3602	0.3668	0.3883	0.3956	0.4066	0.4484	0.4412
DLinear	Base	0.3459	0.3737	0.3818	0.3911	0.4153	0.4150	0.4730	0.4509
	+TimeRAF	0.3205	0.3620	0.3807	0.3898	0.4078	0.4118	0.4557	0.4455
FreTS	Base	0.3398	0.3754	0.3833	0.3984	0.4202	0.4250	0.4975	0.4742
	+TimeRAF	0.3247	0.3663	0.3787	0.3949	0.4159	0.4218	0.4821	0.4670
FEDformer	Base	0.5353	0.4892	0.5273	0.4908	0.6801	0.5477	0.7092	0.5623
	+TimeRAF	0.4252	0.4418	0.4238	0.4314	0.5074	0.4858	0.5556	0.5081
Non-Stat. Transformer	Base	0.5062	0.4644	0.6041	0.4973	0.6640	0.5186	0.7208	0.5431
	+TimeRAF	0.3945	0.4076	0.4652	0.4350	0.5083	0.4560	0.5741	0.4850
LightTS	Base	0.3691	0.3914	0.3994	0.4056	0.4296	0.4261	0.4876	0.4623
	+TimeRAF	0.3384	0.3767	0.3796	0.3967	0.4243	0.4217	0.4682	0.4557
Informer	Base	0.7403	0.6205	0.7609	0.6270	0.8722	0.6878	0.9979	0.7377
	+TimeRAF	0.5049	0.4937	0.5280	0.5034	0.5926	0.5453	0.6693	0.5802
Autoformer	Base	0.5198	0.4821	0.5781	0.5123	0.6100	0.5325	0.6812	0.5671
	+TimeRAF	0.4015	0.4169	0.4486	0.4437	0.4743	0.4629	0.5340	0.4949

Table 21: Complete long-term results on ETTm2. Base and +TimeRAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	96		192		336		720	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TimeXER	Base	0.1712	0.2560	0.2365	0.2987	0.2953	0.3379	0.3934	0.3942
	+TimeRAF	0.1707	0.2557	0.2358	0.2980	0.2926	0.3359	0.3908	0.3924
TimeMixer	Base	0.1750	0.2574	0.2360	0.2975	0.3026	0.3410	0.4009	0.4054
	+TimeRAF	0.1743	0.2569	0.2350	0.2966	0.2994	0.3379	0.3958	0.4021
PATtn	Base	0.1800	0.2668	0.2462	0.3079	0.3084	0.3471	0.4076	0.4031
	+TimeRAF	0.1788	0.2657	0.2448	0.3066	0.3057	0.3451	0.4045	0.4009
iTransformer	Base	0.1862	0.2723	0.2516	0.3119	0.3153	0.3527	0.4132	0.4060
	+TimeRAF	0.1830	0.2658	0.2498	0.3095	0.3125	0.3499	0.4098	0.4031
TimesNet	Base	0.1894	0.2662	0.2634	0.3119	0.3204	0.3488	0.4229	0.4069
	+TimeRAF	0.1807	0.2623	0.2630	0.3114	0.3195	0.3479	0.4208	0.4058
PATCHTST	Base	0.1828	0.2679	0.2463	0.3058	0.3105	0.3492	0.4150	0.4108
	+TimeRAF	0.1793	0.2654	0.2446	0.3045	0.3082	0.3478	0.4108	0.4083
DLinear	Base	0.1934	0.2928	0.2845	0.3614	0.3849	0.4295	0.5561	0.5235
	+TimeRAF	0.1771	0.2639	0.2394	0.3033	0.2840	0.3357	0.3908	0.3993
FreTS	Base	0.1943	0.2846	0.2774	0.3507	0.3698	0.4042	0.5602	0.5186
	+TimeRAF	0.1771	0.2656	0.2593	0.3280	0.2855	0.3384	0.3841	0.3997
FEDformer	Base	0.2098	0.2933	0.2710	0.3288	0.3275	0.3641	0.4204	0.4164
	+TimeRAF	0.1949	0.2845	0.2578	0.3213	0.3186	0.3586	0.4136	0.4051
Non-Stat. Transformer	Base	0.2233	0.3008	0.2680	0.3234	0.4862	0.4538	0.4572	0.4351
	+TimeRAF	0.2029	0.2814	0.2478	0.3067	0.3899	0.3997	0.4260	0.4132
LightTS	Base	0.2088	0.3076	0.3005	0.3748	0.3933	0.4347	0.5449	0.5188
	+TimeRAF	0.1934	0.2831	0.2510	0.3164	0.2983	0.3495	0.3755	0.3993
Informer	Base	0.7444	0.6312	0.9328	0.7436	1.2786	0.8891	1.9379	1.1306
	+TimeRAF	0.2816	0.3771	0.3287	0.4149	0.3936	0.4399	0.4170	0.4590
Autoformer	Base	0.2719	0.3291	0.2845	0.3429	0.3319	0.3669	0.4277	0.4223
	+TimeRAF	0.2212	0.3031	0.2624	0.3277	0.3173	0.3573	0.4162	0.4132

Table 22: Complete long-term results on Weather. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	96		192		336		720	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TIME_XER	Base	0.1786	0.2244	0.2041	0.2475	0.2607	0.2895	0.3400	0.3404
	+TIMERAF	0.1746	0.2236	0.2034	0.2473	0.2598	0.2892	0.3388	0.3398
TIME_MIXER	Base	0.1610	0.2079	0.2064	0.2507	0.2635	0.2917	0.3438	0.3436
	+TIMERAF	0.1606	0.2077	0.2057	0.2503	0.2630	0.2916	0.3423	0.3426
PATTN	Base	0.1894	0.2263	0.2341	0.2641	0.2873	0.3015	0.3622	0.3495
	+TIMERAF	0.1837	0.2257	0.2328	0.2706	0.2862	0.3012	0.3602	0.3488
iTRANSFORMER	Base	0.1971	0.2366	0.2432	0.2738	0.2951	0.3104	0.3611	0.3510
	+TIMERAF	0.1902	0.2340	0.2378	0.2745	0.2879	0.3105	0.3572	0.3490
TIMESNET	Base	0.1690	0.2195	0.2247	0.2640	0.2815	0.3036	0.3566	0.3526
	+TIMERAF	0.1658	0.2168	0.2245	0.2639	0.2811	0.3034	0.3554	0.3519
PATCHTST	Base	0.1722	0.2135	0.2200	0.2556	0.2794	0.2979	0.3546	0.3467
	+TIMERAF	0.1677	0.2129	0.2194	0.2556	0.2788	0.2978	0.3534	0.3462
DLINER	Base	0.1962	0.2561	0.2389	0.2992	0.2811	0.3306	0.3454	0.3819
	+TIMERAF	0.1901	0.2506	0.2360	0.2987	0.2785	0.3308	0.3353	0.3759
FRETS	Base	0.1843	0.2386	0.2226	0.2735	0.2724	0.3175	0.3544	0.3868
	+TIMERAF	0.1797	0.2355	0.2218	0.2771	0.2704	0.3198	0.3519	0.3857
FEDFORMER	Base	0.2713	0.3419	0.3217	0.3774	0.3282	0.3654	0.3938	0.3993
	+TIMERAF	0.2038	0.2778	0.2603	0.3210	0.2997	0.3356	0.3817	0.3959
NON-STAT. TRANSFORMER	Base	0.2086	0.2512	0.2787	0.3057	0.3649	0.3584	0.3928	0.3789
	+TIMERAF	0.1836	0.2317	0.2394	0.2778	0.3159	0.3283	0.3707	0.3636
LIGHTTS	Base	0.1964	0.2552	0.2419	0.2987	0.2899	0.3365	0.3497	0.3834
	+TIMERAF	0.1904	0.2491	0.2392	0.2982	0.2822	0.3323	0.3384	0.3782
INFORMER	Base	0.5204	0.5156	0.5769	0.5391	0.9701	0.7332	1.2664	0.8835
	+TIMERAF	0.3321	0.3994	0.3830	0.4274	0.3254	0.3776	0.3776	0.3929
AUTOFORMER	Base	0.2582	0.3310	0.3039	0.3625	0.3481	0.3888	0.4332	0.4336
	+TIMERAF	0.2056	0.2806	0.2575	0.3186	0.3021	0.3440	0.3907	0.3956

Table 23: Complete long-term results on Electricity. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	96		192		336		720	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TIME_XER	Base	0.1405	0.2425	0.1571	0.2556	0.1769	0.2752	0.2110	0.3061
	+TIMERAF	0.1400	0.2379	0.1564	0.2525	0.1739	0.2693	0.2055	0.3011
TIME_MIXER	Base	0.1562	0.2476	0.1701	0.2610	0.1868	0.2773	0.2283	0.3131
	+TIMERAF	0.1528	0.2447	0.1679	0.2590	0.1841	0.2751	0.2251	0.3105
PATTN	Base	0.1846	0.2668	0.1887	0.2717	0.2043	0.2875	0.2467	0.3216
	+TIMERAF	0.1737	0.2620	0.1865	0.2700	0.2014	0.2854	0.2436	0.3191
iTRANSFORMER	Base	0.1488	0.2406	0.1645	0.2560	0.1784	0.2712	0.2271	0.3114
	+TIMERAF	0.1443	0.2364	0.1611	0.2528	0.1738	0.2670	0.2206	0.3050
TIMESNET	Base	0.1679	0.2716	0.1860	0.2883	0.2036	0.3051	0.2203	0.3183
	+TIMERAF	0.1649	0.2676	0.1739	0.2747	0.1961	0.2948	0.2166	0.3156
PATCHTST	Base	0.1800	0.2730	0.1874	0.2797	0.2043	0.2960	0.2455	0.3283
	+TIMERAF	0.1697	0.2622	0.1835	0.2743	0.1949	0.2856	0.2431	0.3267
DLINER	Base	0.2104	0.3016	0.2102	0.3047	0.2231	0.3192	0.2578	0.3496
	+TIMERAF	0.1897	0.2848	0.2007	0.2950	0.2058	0.3022	0.2551	0.3424
FRETS	Base	0.1895	0.2768	0.1917	0.2792	0.2069	0.2966	0.2467	0.3333
	+TIMERAF	0.1782	0.2704	0.1900	0.2801	0.1987	0.2897	0.2427	0.3270
FEDFORMER	Base	0.1948	0.3089	0.2022	0.3155	0.2292	0.3423	0.2620	0.3655
	+TIMERAF	0.1831	0.2945	0.1905	0.3001	0.2071	0.3188	0.2420	0.3377
NON-STAT. TRANSFORMER	Base	0.8414	0.7579	0.1836	0.2857	0.2002	0.3033	0.8928	0.7727
	+TIMERAF	0.4065	0.4973	0.1705	0.2684	0.1867	0.2887	0.4818	0.5314
LIGHTTS	Base	0.2133	0.3156	0.2218	0.3251	0.2416	0.3445	0.2770	0.3710
	+TIMERAF	0.1955	0.2985	0.2038	0.3065	0.2208	0.3240	0.2560	0.3460
INFORMER	Base	0.3290	0.4131	0.3671	0.4454	0.3523	0.4348	0.4004	0.4597
	+TIMERAF	0.2811	0.3742	0.2954	0.3917	0.2967	0.3923	0.3035	0.3912
AUTOFORMER	Base	0.2006	0.3174	0.2252	0.3333	0.2445	0.3516	0.2763	0.3772
	+TIMERAF	0.1796	0.2865	0.2044	0.3128	0.2250	0.3307	0.2517	0.3525

Table 24: Complete long-term results on Traffic. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	96		192		336		720	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TIMEKER	Base	0.4300	0.2720	0.4477	0.2812	0.4714	0.2889	0.5148	0.3069
	+TIMERAF	0.4247	0.2679	0.4416	0.2761	0.4647	0.2843	0.5086	0.3029
TIMEMIXER	Base	0.4743	0.2963	0.5029	0.3016	0.5281	0.3083	0.5543	0.3233
	+TIMERAF	0.4618	0.2853	0.4941	0.2922	0.5180	0.2973	0.5446	0.3138
PATTN	Base	0.5050	0.3298	0.5023	0.3234	0.5153	0.3275	0.5489	0.3459
	+TIMERAF	0.4876	0.3169	0.4907	0.3140	0.5028	0.3180	0.5361	0.3366
iTRANSFORMER	Base	0.3936	0.2686	0.4123	0.2770	0.4235	0.2828	0.4581	0.3002
	+TIMERAF	0.3786	0.2599	0.4057	0.2701	0.4163	0.2760	0.4510	0.2938
TIMESNET	Base	0.5900	0.3144	0.6147	0.3243	0.6347	0.3380	0.6557	0.3467
	+TIMERAF	0.5302	0.3041	0.5801	0.3189	0.6289	0.3338	0.6231	0.3422
PATCHTST	Base	0.4582	0.2981	0.4687	0.3008	0.4828	0.3074	0.5174	0.3261
	+TIMERAF	0.4307	0.2827	0.4619	0.2958	0.4752	0.3024	0.5097	0.3215
DLINER	Base	0.6965	0.4287	0.6466	0.4074	0.6532	0.4097	0.6940	0.4286
	+TIMERAF	0.5378	0.3537	0.5234	0.3452	0.5719	0.3708	0.6727	0.4188
FRETS	Base	0.5649	0.3679	0.5683	0.3650	0.6004	0.3737	0.6614	0.3978
	+TIMERAF	0.5374	0.3453	0.5467	0.3449	0.5781	0.3543	0.6410	0.3808
FEDFORMER	Base	0.5775	0.3624	0.6095	0.3800	0.6266	0.3901	0.6511	0.3988
	+TIMERAF	0.5514	0.3487	0.5744	0.3591	0.5935	0.3730	0.6258	0.3862
NON-STAT. TRANSFORMER	Base	0.6307	0.3409	0.6371	0.3457	0.6565	0.3495	0.6774	0.3631
	+TIMERAF	0.5687	0.3301	0.5798	0.3328	0.6164	0.3411	0.6401	0.3549
LIGHTTS	Base	0.6812	0.4218	0.6701	0.4270	0.6897	0.4360	0.7443	0.4561
	+TIMERAF	0.6264	0.3803	0.6160	0.3854	0.6346	0.3947	0.6896	0.4169
INFORMER	Base	0.7498	0.4238	0.8101	0.4606	0.8660	0.4898	1.0016	0.5617
	+TIMERAF	0.6605	0.3888	0.6807	0.4098	0.7350	0.4374	0.8322	0.4921
AUTOFORMER	Base	0.6726	0.4027	0.6330	0.3969	0.6476	0.4041	0.6688	0.4133
	+TIMERAF	0.6262	0.3818	0.5847	0.3689	0.6012	0.3821	0.6295	0.3924

Table 25: Complete PEMS results on PEMS03. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	6			12			24		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
TIMEKER	Base	16.559	26.203	0.154	18.376	28.527	0.176	21.692	33.321	0.216
	+TIMERAF	14.533	22.930	0.137	16.004	25.151	0.153	18.687	29.249	0.185
TIMEMIXER	Base	13.910	21.763	0.138	15.355	24.195	0.155	17.252	27.246	0.176
	+TIMERAF	13.682	21.524	0.135	14.928	23.702	0.148	16.532	26.410	0.165
PATTN	Base	15.733	24.870	0.145	18.943	30.139	0.175	24.975	40.034	0.231
	+TIMERAF	14.506	22.904	0.136	16.694	26.378	0.158	20.900	32.963	0.201
iTRANSFORMER	Base	15.031	23.749	0.140	17.389	27.595	0.163	21.992	35.361	0.213
	+TIMERAF	14.193	22.464	0.132	15.957	25.289	0.151	19.354	31.072	0.188
TIMESNET	Base	16.513	26.352	0.164	17.732	28.254	0.177	20.112	32.264	0.202
	+TIMERAF	15.725	25.025	0.154	16.969	27.382	0.168	18.993	30.957	0.190
PATCHTST	Base	16.619	25.762	0.163	20.246	31.971	0.180	25.350	40.365	0.227
	+TIMERAF	14.641	22.984	0.142	16.798	26.499	0.156	20.601	32.524	0.194
DLINER	Base	18.971	29.982	0.195	22.877	36.186	0.236	29.958	47.391	0.307
	+TIMERAF	15.395	24.224	0.156	17.668	27.850	0.180	21.654	34.106	0.222
FRETS	Base	15.382	24.357	0.143	17.898	28.408	0.169	22.465	35.778	0.215
	+TIMERAF	14.272	22.523	0.134	15.962	25.304	0.153	19.138	30.402	0.186
FEDFORMER	Base	25.876	37.318	0.294	26.304	38.518	0.278	29.555	43.522	0.301
	+TIMERAF	21.596	32.477	0.241	21.652	33.021	0.229	24.338	36.877	0.253
NON-STAT. TRANSFORMER	Base	16.294	26.154	0.161	17.917	28.759	0.177	20.669	33.785	0.202
	+TIMERAF	15.501	24.778	0.151	16.992	27.721	0.166	19.511	32.409	0.189
LIGHTTS	Base	17.656	27.076	0.180	18.792	29.051	0.189	21.942	33.751	0.218
	+TIMERAF	16.505	25.565	0.169	17.235	26.777	0.176	19.705	30.666	0.199
INFORMER	Base	18.424	31.355	0.186	19.623	32.938	0.194	20.855	34.355	0.208
	+TIMERAF	15.962	25.736	0.157	17.242	27.509	0.166	19.482	32.970	0.191
AUTOFORMER	Base	33.587	50.426	0.366	29.025	42.609	0.290	35.903	52.458	0.384
	+TIMERAF	26.409	39.679	0.290	23.989	35.783	0.252	29.511	43.494	0.324

Table 26: Complete PEMS results on PEMS04. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	6			12			24		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
TIMEXER	Base	22.215	33.451	0.158	24.422	36.491	0.172	28.035	42.229	0.188
	+TIMERAF	19.773	31.039	0.131	21.278	33.117	0.141	24.193	37.410	0.157
TIMEMIXER	Base	18.747	29.822	0.121	19.750	31.202	0.130	21.490	33.618	0.142
	+TIMERAF	20.598	31.501	0.138	23.184	34.212	0.172	25.496	37.059	0.194
PATTN	Base	21.504	33.940	0.137	25.486	39.844	0.161	33.920	52.384	0.219
	+TIMERAF	20.033	32.110	0.126	22.544	35.754	0.142	27.762	42.966	0.181
tTRANSFORMER	Base	20.731	32.867	0.132	23.640	37.288	0.150	29.372	45.862	0.193
	+TIMERAF	19.384	31.277	0.121	21.307	34.101	0.133	25.020	39.300	0.161
TIMESNET	Base	21.198	33.376	0.139	22.399	34.951	0.149	24.460	38.015	0.165
	+TIMERAF	20.691	32.576	0.133	21.639	34.162	0.142	23.186	36.594	0.154
PATCHTST	Base	23.341	35.083	0.164	26.846	40.585	0.182	34.293	52.480	0.220
	+TIMERAF	20.428	32.074	0.135	22.683	35.464	0.148	27.324	42.093	0.177
DLINER	Base	27.263	40.283	0.213	32.003	46.974	0.254	40.203	58.240	0.329
	+TIMERAF	21.705	33.227	0.156	24.145	36.439	0.178	28.285	41.726	0.219
FRETS	Base	21.054	33.114	0.138	24.229	37.670	0.159	30.438	46.446	0.207
	+TIMERAF	19.620	31.434	0.127	21.601	34.279	0.139	25.284	39.138	0.169
FEDFORMER	Base	32.008	45.741	0.254	32.490	47.083	0.251	34.952	51.084	0.268
	+TIMERAF	29.034	42.324	0.213	27.308	41.068	0.200	29.315	44.502	0.215
NON-STAT. TRANSFORMER	Base	20.983	33.107	0.138	21.942	34.550	0.143	23.828	37.152	0.160
	+TIMERAF	20.548	32.465	0.130	21.264	33.836	0.138	22.937	36.120	0.153
LIGHTTS	Base	24.701	37.135	0.174	25.326	38.118	0.179	29.292	43.597	0.210
	+TIMERAF	22.921	35.024	0.161	23.379	35.888	0.163	25.998	39.258	0.185
INFORMER	Base	22.615	36.234	0.149	23.238	36.965	0.153	24.321	38.028	0.166
	+TIMERAF	21.122	33.212	0.137	21.633	33.746	0.141	23.474	35.975	0.159
AUTOFORMER	Base	33.137	47.924	0.247	45.172	63.391	0.374	54.568	76.742	0.410
	+TIMERAF	28.475	41.736	0.198	36.502	51.311	0.307	41.372	58.130	0.323

Table 27: Complete PEMS results on PEMS07. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	6			12			24		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
TIMEXER	Base	24.304	35.922	0.109	27.061	39.855	0.126	31.441	46.343	0.148
	+TIMERAF	21.117	32.784	0.090	23.108	35.720	0.101	26.407	40.569	0.118
TIMEMIXER	Base	19.752	31.426	0.083	21.294	33.866	0.090	23.887	37.663	0.104
	+TIMERAF	20.324	31.797	0.086	22.430	34.635	0.099	22.836	36.460	0.098
PATTN	Base	23.070	35.858	0.096	27.835	42.820	0.120	37.961	57.703	0.173
	+TIMERAF	21.089	33.602	0.087	24.232	38.267	0.103	30.524	47.089	0.139
tTRANSFORMER	Base	21.906	34.509	0.092	25.139	39.516	0.107	31.897	49.474	0.141
	+TIMERAF	20.450	32.793	0.084	22.712	36.375	0.095	27.079	42.581	0.118
TIMESNET	Base	25.050	39.946	0.108	26.321	41.912	0.116	29.188	46.055	0.133
	+TIMERAF	23.416	37.346	0.098	24.646	39.870	0.107	26.875	43.205	0.120
PATCHTST	Base	25.770	38.135	0.110	28.676	43.199	0.132	38.353	57.110	0.181
	+TIMERAF	21.906	33.940	0.091	24.241	37.911	0.107	30.177	46.248	0.138
DLINER	Base	27.854	40.792	0.136	34.096	49.306	0.175	46.074	65.728	0.250
	+TIMERAF	22.562	34.471	0.103	26.041	39.075	0.125	32.371	47.135	0.167
FRETS	Base	22.382	34.921	0.093	26.479	40.803	0.116	34.553	52.449	0.156
	+TIMERAF	20.637	33.012	0.085	23.349	37.077	0.100	28.214	43.822	0.126
FEDFORMER	Base	36.665	51.492	0.188	36.897	52.679	0.184	36.887	53.140	0.186
	+TIMERAF	33.124	46.800	0.162	31.458	46.439	0.151	31.577	46.910	0.154
NON-STAT. TRANSFORMER	Base	24.985	39.798	0.109	26.230	41.803	0.115	28.998	45.932	0.134
	+TIMERAF	23.496	37.385	0.099	24.641	39.964	0.107	26.888	43.480	0.121
LIGHTTS	Base	25.362	38.053	0.115	26.393	39.420	0.120	30.613	45.129	0.145
	+TIMERAF	23.650	36.303	0.105	24.448	37.438	0.111	27.333	41.394	0.128
INFORMER	Base	28.279	45.443	0.122	29.154	46.374	0.126	30.464	47.648	0.135
	+TIMERAF	25.251	39.735	0.105	25.374	39.274	0.109	28.311	43.308	0.127
AUTOFORMER	Base	42.768	59.916	0.229	43.315	63.301	0.195	48.308	66.696	0.257
	+TIMERAF	39.267	55.458	0.189	36.169	53.193	0.166	39.815	56.115	0.207

Table 28: Complete PEMS results on PEMS08. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	6			12			24		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
TIMEXER	Base	18.683	28.645	0.114	20.919	31.487	0.134	25.861	39.061	0.164
	+TIMERAF	15.945	24.634	0.099	17.516	26.798	0.113	21.184	32.267	0.138
TIMEMIXER	Base	14.559	22.851	0.092	15.940	24.773	0.102	17.957	28.006	0.117
	+TIMERAF	14.139	22.341	0.089	15.322	24.040	0.098	16.952	26.798	0.110
PATTN	Base	17.381	27.464	0.102	20.291	31.935	0.121	27.034	42.130	0.165
	+TIMERAF	15.658	24.680	0.095	17.625	27.695	0.109	21.936	33.965	0.139
iTRANSFORMER	Base	16.364	26.167	0.099	18.688	29.602	0.115	24.182	37.991	0.151
	+TIMERAF	14.962	23.869	0.092	16.603	26.280	0.104	20.339	31.662	0.130
TIMESNET	Base	18.981	29.403	0.116	20.590	31.807	0.130	23.116	35.217	0.148
	+TIMERAF	17.780	27.920	0.107	19.193	29.822	0.118	20.768	32.310	0.136
PATCHTST	Base	18.967	29.108	0.113	21.668	33.094	0.134	28.246	43.562	0.173
	+TIMERAF	16.092	24.957	0.099	17.859	27.672	0.113	22.221	34.265	0.143
DLINER	Base	23.102	34.846	0.144	26.421	39.072	0.171	33.929	49.463	0.226
	+TIMERAF	17.672	26.887	0.110	19.469	29.175	0.125	23.583	34.702	0.156
FRETS	Base	16.453	26.153	0.099	18.799	29.610	0.117	23.813	37.008	0.151
	+TIMERAF	15.062	23.889	0.093	16.545	26.163	0.105	19.711	30.720	0.127
FEDFORMER	Base	26.277	37.762	0.183	26.718	38.576	0.182	29.564	43.005	0.203
	+TIMERAF	24.537	36.594	0.152	24.066	35.410	0.156	23.868	35.615	0.164
NON-STAT. TRANSFORMER	Base	17.464	27.001	0.108	18.808	28.916	0.120	21.376	33.074	0.138
	+TIMERAF	16.874	26.407	0.103	18.044	28.022	0.112	19.762	31.002	0.130
LIGHTTS	Base	19.279	29.497	0.119	21.080	31.857	0.137	24.081	36.107	0.156
	+TIMERAF	17.299	26.469	0.111	18.492	28.164	0.122	20.521	30.996	0.135
INFORMER	Base	21.383	32.541	0.138	22.577	34.086	0.147	23.819	35.744	0.156
	+TIMERAF	19.577	30.630	0.119	20.624	31.303	0.130	22.901	34.009	0.148
AUTOFORMER	Base	31.425	44.967	0.195	38.215	54.727	0.226	50.192	73.587	0.317
	+TIMERAF	29.328	42.059	0.180	29.748	42.912	0.182	38.095	54.786	0.249

Table 29: Complete EPF results on NP. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	MSE	MAE
TIMEXER	Base	0.2434	0.2714
	+TIMERAF	0.2498	0.2733
TIMEMIXER	Base	0.2642	0.2899
	+TIMERAF	0.2612	0.2841
PATTN	Base	0.2774	0.3018
	+TIMERAF	0.2700	0.2926
iTRANSFORMER	Base	0.2599	0.2875
	+TIMERAF	0.2580	0.2807
TIMESNET	Base	0.2432	0.2848
	+TIMERAF	0.2403	0.2751
PATCHTST	Base	0.2814	0.2991
	+TIMERAF	0.2781	0.2940
DLINER	Base	0.2966	0.3115
	+TIMERAF	0.2919	0.3091
FRETS	Base	0.3105	0.3215
	+TIMERAF	0.3049	0.3184
FEDFORMER	Base	0.3284	0.3641
	+TIMERAF	0.2973	0.3266
NON-STAT. TRANSFORMER	Base	0.5862	0.4916
	+TIMERAF	0.4833	0.4267
LIGHTTS	Base	0.2991	0.3178
	+TIMERAF	0.2919	0.3082
INFORMER	Base	0.3037	0.3486
	+TIMERAF	0.2747	0.3180
AUTOFORMER	Base	0.3823	0.3809
	+TIMERAF	0.3284	0.3402

Table 30: Complete EPF results on PJM. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	MSE	MAE
TIMEKER	Base	0.0968	0.1944
	+TIMERAF	0.0952	0.1930
TIMEMIXER	Base	0.1176	0.2302
	+TIMERAF	0.1095	0.2174
PATTN	Base	0.1114	0.2160
	+TIMERAF	0.1089	0.2111
ITRANSFORMER	Base	0.0960	0.1958
	+TIMERAF	0.0925	0.1928
TIMESNET	Base	0.1239	0.2285
	+TIMERAF	0.0998	0.2037
PATCHTST	Base	0.1077	0.2129
	+TIMERAF	0.1031	0.2065
DLINER	Base	0.1042	0.2110
	+TIMERAF	0.0954	0.1995
FRETTS	Base	0.1218	0.2305
	+TIMERAF	0.1121	0.2156
FEDFORMER	Base	0.1312	0.2371
	+TIMERAF	0.1079	0.2110
NON-STAT. TRANSFORMER	Base	0.2582	0.3609
	+TIMERAF	0.1726	0.2737
LIGHTTS	Base	0.1092	0.2119
	+TIMERAF	0.1074	0.2075
INFORMER	Base	0.1287	0.2128
	+TIMERAF	0.1001	0.1953
AUTOFORMER	Base	0.1613	0.2662
	+TIMERAF	0.1216	0.2283

Table 31: Complete EPF results on BE. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	MSE	MAE
TIMEKER	Base	0.3794	0.2423
	+TIMERAF	0.3716	0.2414
TIMEMIXER	Base	0.4146	0.2758
	+TIMERAF	0.4135	0.2746
PATTN	Base	0.3998	0.2661
	+TIMERAF	0.3959	0.2653
ITRANSFORMER	Base	0.4042	0.2767
	+TIMERAF	0.4103	0.2777
TIMESNET	Base	0.3899	0.2696
	+TIMERAF	0.3881	0.2684
PATCHTST	Base	0.4041	0.2610
	+TIMERAF	0.3944	0.2592
DLINER	Base	0.4591	0.3147
	+TIMERAF	0.4351	0.3084
FRETTS	Base	0.4370	0.3106
	+TIMERAF	0.4240	0.2988
FEDFORMER	Base	0.4739	0.3324
	+TIMERAF	0.4157	0.3044
NON-STAT. TRANSFORMER	Base	0.6884	0.4620
	+TIMERAF	0.6070	0.3804
LIGHTTS	Base	0.4436	0.2967
	+TIMERAF	0.4268	0.2864
INFORMER	Base	0.4841	0.3247
	+TIMERAF	0.4417	0.3130
AUTOFORMER	Base	0.5049	0.3219
	+TIMERAF	0.4039	0.2862

Table 32: Complete EPF results on FR. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	MSE	MAE
TIMEKER	Base	0.3960	0.2119
	+TIMERAF	0.3926	0.2069
TIMEMIXER	Base	0.4445	0.2447
	+TIMERAF	0.4047	0.2332
PATTN	Base	0.4692	0.2602
	+TIMERAF	0.4604	0.2510
ITRANSFORMER	Base	0.4203	0.2279
	+TIMERAF	0.4170	0.2271
TIMESNET	Base	0.4233	0.2295
	+TIMERAF	0.4113	0.2259
PATCHTST	Base	0.4135	0.2236
	+TIMERAF	0.4087	0.2156
DLINER	Base	0.4196	0.2537
	+TIMERAF	0.3943	0.2225
FRETTS	Base	0.4030	0.2356
	+TIMERAF	0.3872	0.2243
FEDFORMER	Base	0.4563	0.2762
	+TIMERAF	0.3969	0.2460
NON-STAT. TRANSFORMER	Base	0.6214	0.3883
	+TIMERAF	0.5330	0.2953
LIGHTTS	Base	0.4734	0.2581
	+TIMERAF	0.4691	0.2506
INFORMER	Base	0.4785	0.2680
	+TIMERAF	0.4645	0.2507
AUTOFORMER	Base	0.6188	0.3017
	+TIMERAF	0.4783	0.2608

Table 33: Complete EPF results on DE. Base and +TIMERAF are reported separately for every backbone and horizon; bold marks a lower corrected error. No result in this table is averaged over backbones or horizons.

Backbone	System	MSE	MAE
TIMEKER	Base	0.4581	0.4224
	+TIMERAF	0.4610	0.4258
TIMEMIXER	Base	0.4690	0.4370
	+TIMERAF	0.4736	0.4371
PATTN	Base	0.4676	0.4348
	+TIMERAF	0.4693	0.4353
ITRANSFORMER	Base	0.4660	0.4353
	+TIMERAF	0.4580	0.4323
TIMESNET	Base	0.4747	0.4331
	+TIMERAF	0.4752	0.4332
PATCHTST	Base	0.5099	0.4521
	+TIMERAF	0.5071	0.4507
DLINER	Base	0.5081	0.4567
	+TIMERAF	0.4970	0.4433
FRETTS	Base	0.5109	0.4517
	+TIMERAF	0.5149	0.4524
FEDFORMER	Base	0.6164	0.5268
	+TIMERAF	0.5731	0.4955
NON-STAT. TRANSFORMER	Base	0.5171	0.4599
	+TIMERAF	0.5258	0.4700
LIGHTTS	Base	0.4891	0.4421
	+TIMERAF	0.4931	0.4433
INFORMER	Base	0.5849	0.4930
	+TIMERAF	0.5568	0.4764
AUTOFORMER	Base	0.5967	0.5069
	+TIMERAF	0.5606	0.4871

G ADDITIONAL FORECASTING SYSTEMS

Table 35 gives a compact view of every dataset–system cell. The entry is the smallest reduction among the metrics reported for that cell, so positive values correspond exactly to the strict criterion used in the main text. Each static ensemble is one forecasting system, and each AUTOGLUON or CHRONOS-BOLT variant is one trained forecaster. The accompanying machine-readable results retain the absolute metrics.

Table 34: **Correction of additional ensemble, AutoML, and foundation-model systems.** Each system is evaluated once per dataset at horizon 96 (long-term) or 24 (PEMS and EPF). “Strict” requires every applicable metric to improve. Entries Δ are mean paired error reductions in percent (higher is better). MSE uses 12 cells per system except AutoGluon (10); MAE uses 16 except AutoGluon (14); RMSE and MAPE use four.

System	Eval.	Strict	Rate	Δ MSE	Δ MAE	Δ RMSE	Δ MAPE
FORWARD SELECTION	16	13	81.2%	0.87	1.56	3.12	5.35
PORTFOLIO ENSEMBLE	16	12	75.0%	1.17	1.57	3.06	5.62
ZEROSHOT ENSEMBLE	16	13	81.2%	1.40	2.68	5.44	8.01
AUTOGLUON HQ	14	13	92.9%	13.10	14.87	27.32	28.88
CHRONOS-BOLT FT	16	14	87.5%	8.85	8.61	24.42	21.62
CHRONOS-BOLT ZS	16	16	100.0%	10.74	9.36	26.97	25.50
All systems	94	81	86.2%	5.82	6.26	15.05	15.83

Table 35: **Dataset-level robustness for additional systems.** Each entry is the smallest paired error reduction across the metrics reported for that cell; it is positive exactly when the cell is a strict win. Higher is better.

Family	Dataset	FORWARD SELECTION	PORTFOLIO ENSEMBLE	ZEROSHOT ENSEMBLE	AUTOGLUON HQ	CHRONOS-BOLT FT	CHRONOS-BOLT ZS
Long-term	ETTh1	0.03	-0.20	0.89	6.69	2.25	1.74
	ETTh2	-0.12	-0.24	-0.35	2.79	2.56	1.41
	ETTm1	1.15	1.35	1.85	13.45	13.86	11.51
	ETTm2	0.72	0.64	0.91	7.41	5.37	7.44
	weather	0.55	-0.14	-0.03	5.75	9.34	10.75
	electricity	-0.04	0.71	1.41	OOT	1.94	1.01
	traffic	1.79	2.71	4.79	OOT	7.36	2.94
PEMS	PEMS03	2.55	2.51	4.52	26.91	19.49	24.56
	PEMS04	1.79	1.82	3.98	24.81	19.47	24.94
	PEMS07	2.91	2.64	5.13	27.03	23.55	26.12
	PEMS08	5.24	5.28	8.13	28.39	23.10	25.13
EPF	NP	-0.78	-0.69	-1.01	17.44	0.85	0.48
	PJM	0.36	0.43	0.42	20.13	0.77	0.61
	BE	0.38	0.90	0.47	-6.01	-0.22	0.33
	FR	0.04	0.21	0.33	4.71	0.03	0.66
	DE	0.11	0.55	0.20	9.11	-1.04	4.05

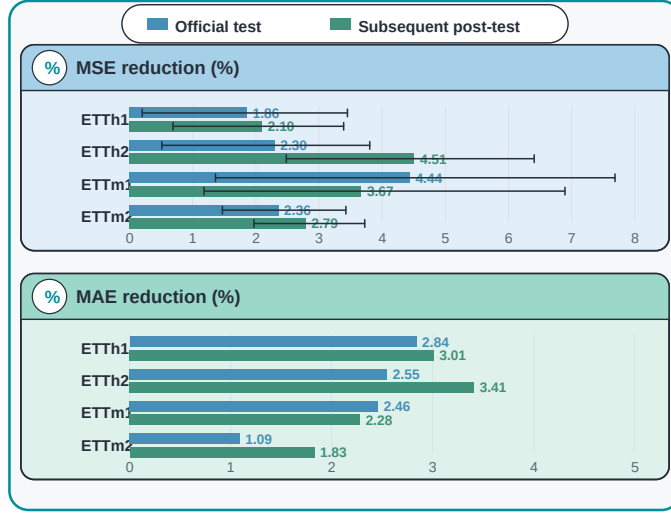


Figure 6: Pooled MSE and MAE reductions over matched TIMEMIXER forecasts across three seeds. The post-test timeline is chronologically subsequent to the standard split and is excluded from model and correction selection. Horizontal lines on MSE bars are 95% paired moving-block-bootstrap intervals. Higher is better.

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Table 36: Pooled MSE reductions and 95% paired moving-block-bootstrap intervals. Intervals are candidate-minus-baseline MSE, so values below zero favor TIMERAf.

Dataset	Official test		Subsequent post-test	
	Reduction	95% interval	Reduction	95% interval
ETTh1	1.86%	$[-0.013169, -0.000768]$	2.10%	$[-0.017964, -0.003646]$
ETTh2	2.30%	$[-0.011055, -0.001485]$	4.51%	$[-0.011830, -0.004583]$
ETTm1	4.44%	$[-0.024903, -0.004411]$	3.67%	$[-0.034576, -0.005918]$
ETTm2	2.36%	$[-0.006030, -0.002587]$	2.79%	$[-0.004747, -0.002511]$

Table 37: Per-seed MSE reductions (%) for the ETT prequential study. All corresponding MAE values also improve.

Dataset	Official test			Subsequent post-test		
	2021	2022	2023	2021	2022	2023
ETTh1	1.59	1.49	2.48	2.39	2.08	1.84
ETTh2	4.31	1.18	1.37	6.29	3.69	3.49
ETTm1	2.72	8.35	2.06	2.12	5.50	3.39
ETTm2	2.36	2.54	2.17	2.62	2.72	3.02

The intervals use 5,000 moving-block bootstrap resamples of length 96 after averaging aligned seed losses at each origin. The later timeline begins on 2018-02-21 for ETTh and ends on 2018-06-25 for ETTm. ETTh uses the fixed stack specified in the experiment protocol; ETTm retrieval adds an official-test residual only after its target interval has elapsed.